Distributed Primal-Dual Algorithm for Constrained Multi-agent Reinforcement Learning under Coupled Policies

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***Abstract*—This paper investigates constrained multi-agent reinforcement learning (CMARL) in coupled environments, where agents collaboratively maximize the sum of local objectives while satisfying individual safety constraints. Existing studies face two limitations: (1) most rely on independent policies that fail to capture complex interactions in coupled environments; and (2) agents require the global Lagrange multipliers, which are sensitive learned variables whose global sharing risks exposing private agent-specific information. To overcome these issues, we propose a framework where agents adopt coupled policies that depend on both the states and policy parameters of their κ_p-hop neighbors, where $\kappa_p > 0$ denotes the coupling distance, and develop a distributed and scalable primal-dual (DSPD) algorithm wherein each agent accesses only information within a prescribed local neighborhood. In the proposed algorithm, agents exchange sensitive parameters only with immediate neighbors over a separate time-varying network, while maintaining local estimates to execute the coupled policy. We establish that the proposed algorithm achieves ϵ-policy stationary convergence with approximation error $\mathcal{O}(\gamma^{\frac{\kappa+1}{\kappa_p}})$, where $\kappa > 0$ is the truncated distance and $\gamma \in (0,1)$ is discount factor. Simulations on a wireless access-control network demonstrate that the proposed algorithm outperforms existing state-of-the-art algorithms, validating its effectiveness.**

***Index Terms*—Constrained multi-agent reinforcement learning, coupled policies, distributed primal-dual algorithm, ϵ-policy stationary convergence.**

I. Introduction

Reinforcement learning (RL) [1], [2] has achieved considerable success across a broad spectrum of decision-making applications. However, many real-world systems impose stringent safety requirements, wherein agent must maximize the objective performance while ensuring compliance with safety constraints. For example, in wireless access networks [3], transmitting agents must collaboratively maximize packet transmission throughput subject to individual transmission budget constraints. These safety-critical applications have spurred extensive research on constrained reinforcement learning (CRL) [4]–[8]. For single-agent CRL, a widely adopted approach is provided by Lagrangian-based algorithms [9]–[11], where the primal variable is updated through policy gradient ascent and the dual variable is adjusted using projected subgradient descent.

Extending CRL to multi-agent environments gives rise to constrained multi-agent reinforcement learning (CMARL), which faces two fundamental challenges: i) the curse of dimensionality arising from the exponentially growing joint state-action space, and ii) achieving cooperative optimization while satisfying individual safety constraints. We note that cooperative CMARL constitutes an identical-interest game and therefore falls within the class of Markov potential games (MPGs). Recent studies on (constrained) MPGs [12]–[14] have established convergence to Nash equilibria. However, these methods require every agent to access the global state, resulting in the curse of dimensionality and poor scalability in large-scale networks. This limitation also persists in existing decentralized multi-agent RL algorithms [15], [16], which likewise assume global state access. Although distributed gradient descent and consensus-based methods in distributed optimization [17], [18] share conceptual similarities with CMARL, their analyses cannot be directly extended to the present setting. In particular, both the objective and constraint functions must be estimated from sampled trajectories, the resulting optimization problem is generally non-convex and therefore admits at best convergence to first-order stationary solutions. To address these challenges while ensuring scalability, [19] proposed a truncation-based approach exploiting the exponential decay property, and [3] extended this to scalable CMARL using a primal-dual actor-critic algorithm with local neighborhood information.

Despite these advancements, [3] presents two key limitations. First, its parameter-independent policy architecture fails to capture the inter-agent dependencies arising from coupled state-action dynamics in safety-critical applications such as intelligent transportation systems [20]–[22]. Second, [3] requires agents to access the global Lagrange multipliers, introducing the risk of sensitive information exposure. These limitations inspire our investigation of CMARL under coupled policies, where each agent's policy depends on the states and parameters of its κ_p-hop neighbors, with private information (e.g., policy parameters and Lagrange multipliers) restricted to a separate time-varying network. This design introduces three fundamental challenges. (1) Collaboratively optimizing under coupled policies while satisfying individual safety constraints is significantly more challenging than the independent policy setting; (2) Restricting the exchange of sensitive policy parameters and Lagrange multipliers to local neighborhoods while guaranteeing convergence is nontrivial; and (3) Existing analytical frameworks for independent policies are inapplicable,

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necessitating new theoretical techniques.

To address these challenges, the main contributions are as follows.

1) A novel approximated coupled policy gradient is proposed, which is directly constructed from reward information within the $(\kappa+2\kappa_p)$-hop neighborhood and state-action pairs within the $2\kappa_p$-hop neighborhood, where $\kappa > 0$ is the truncation distance and $\kappa_p > 0$ is the coupling distance, eliminating the need for global state-action information. A geometric 2-horizon sampling scheme is employed to provide an unbiased estimate of the approximated coupled policy gradient and the Lagrangian gradient, enabling coordinated policy updates while satisfying local safety constraints.
2) To reduce the exposure of sensitive agent-specific information, each agent maintains local estimates of other agents' policy parameters and Lagrange multipliers, while exchanging true variables only with immediate neighbors over time-varying learning networks. A distributed update mechanism based on the push-sum protocol is developed for these local estimates, which not only refines each agent's understanding of others' parameters but also promotes the generation of informative samples that guide coordinated policy improvement. Building upon the above two designs, a distributed and scalable primal-dual (DSPD) algorithm is proposed, where each agent operates solely based on information from a prescribed local neighborhood without requiring access to global information.
3) A new theoretical analysis framework is developed for the proposed DSPD algorithm under coupled policies. Specifically, we establish: (i) an approximation error bound for the coupled policy gradient involving both the truncation distance κ and the coupling distance κ_p (see Theorem 1); (ii) the convergence of both policy parameter and Lagrange multiplier estimates to their true values via a carefully ordered interleaved analysis that resolves the circular dependency between the two estimation processes (see Theorem 2); and (iii) an ϵ-policy stationary convergence guarantee with approximation error $\mathcal{O}(\gamma^{\frac{\kappa+1}{\kappa_p}})$, where $\gamma \in (0,1)$ is discount factor (see Theorem 3).

The rest of this paper is organized as follows. Section II introduces network preliminaries. Section III formulates the CMARL problem and derives the coupled policy gradient. Section IV develops the DSPD algorithm. Section V establishes the ϵ-policy stationary convergence guarantee of the DSPD algorithm. Section VI validates the proposed algorithm on a wireless access-control network. Section VII concludes the paper.

Notations: Let $\mathbb{N}$, $\mathbb{R}$, and $\mathbb{R}_+$ denote the sets of natural, real, and positive real numbers, respectively. For $N \in \mathbb{N}$, $\mathbb{R}^N$ ($\mathbb{R}_+^N$) denotes the N-dimensional Euclidean space (positive orthant), and $\mathbf{0}_N$ denotes the zero vector. For $a, b \in \mathbb{R}$, $\mathrm{mod}(a,b)$ and $\lfloor a \rfloor$ denote the modulo and floor operations, respectively. For $x \in \mathbb{R}^N$, $\|x\|$ denotes its Euclidean norm. $(\cdot)^\top$ and $\mathbb{E}[\cdot]$ denote the transpose and expectation operators, respectively.

II. Network preliminaries

Let $\mathcal{G}(\mathcal{N}, \mathcal{E})$ represent a directed network, where $\mathcal{N} = \{1, \cdots, N\}$ denotes the set of agents, $\mathcal{E}$ represents the set of directed edges, and $e_{ij} \in \mathcal{E}$ indicates that agent i can receive information from agent j. $\mathcal{G}$ is strongly connected if there exists at least one directed path from any agent $j \in \mathcal{N}$ to every other agent $i \in \mathcal{N}$. Define $\mathcal{N}_i = \{j | e_{ij} \in \mathcal{E}\}$ as the neighborhood of agent i (including i itself), and $\mathcal{N}_i^\kappa$ as the κ-hop neighborhood, i.e., agents within graph distance κ from agent i (including i itself). Let $\mathcal{N}_{i,-j}^\kappa = \mathcal{N}_i^\kappa \setminus \{j\}$ and $\mathcal{N}_{-i}^\kappa = \mathcal{N} \setminus \mathcal{N}_i^\kappa$.

Let $\{\mathcal{G}_m(\mathcal{N}, \mathcal{E}_m)\}_{m \geq 1}$ represent time-varying networks, which are uniformly strongly connected if there exists an integer $D > 0$ such that $(\mathcal{N}, \bigcup_{m=kD+1}^{(k+1)D} \mathcal{E}_m)$ is strongly connected for any $k \in \mathbb{N}$. For $\mathcal{G}_m$, denote $\mathcal{N}_{i,m}^{out} = \{j | e_{ji,m} \in \mathcal{E}_m, \forall j \in \mathcal{N}\}$ as the out-neighborhood of agent i, and $W_m = [w_{ij,m}]_{N \times N}$ as the weight matrix with

$$w_{ij,m} = \begin{cases} \frac{1}{|\mathcal{N}_{j,m}^{out}|}, & \text{if } i \in \mathcal{N}_{j,m}^{out} \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

where $|\mathcal{N}_{j,m}^{out}|$ is the number of agents in agent j's out-neighborhood.

III. CMARL problem under coupled policies

In this section, we formulate the CMARL problem under coupled policies, define the ϵ-FOSP, and derive the coupled policy gradient.

A. Problem formulation

The CMARL problem under coupled policies is formally defined as $\big(\mathcal{G}^E(\mathcal{N}, \mathcal{E}^E), \{\mathcal{S}_i\}_{i\in\mathcal{N}}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, \{\mathcal{P}_i\}_{i\in\mathcal{N}}, \{f_i\}_{i\in\mathcal{N}}, \{g_i\}_{i\in\mathcal{N}}, \{\pi_i\}_{i\in\mathcal{N}}, \boldsymbol{\rho}, \gamma\big)$, where each element is summarized below.

Environmental network: $\mathcal{G}^E\big(\mathcal{N}, \mathcal{E}^E\big)$ is an environmental network among agents, where $\mathcal{N} = \{1, \cdots, N\}$ denotes the set of agents and the edge set $\mathcal{E}^E$ specifies the links between them. We employ the superscript "E" for the corresponding concepts in $\mathcal{G}^E$.

State and action: $s_i \in \mathcal{S}_i$ and $a_i \in \mathcal{A}_i$ denote the local state and action of agent i. The global state and action are $\boldsymbol{s} = (s_1, \cdots, s_N) \in \boldsymbol{\mathcal{S}} = \prod_{i=1}^N \mathcal{S}_i$ and $\boldsymbol{a} = (a_1, \cdots, a_N) \in \boldsymbol{\mathcal{A}} = \prod_{i=1}^N \mathcal{A}_i$, and $s_{\mathcal{N}_i^{E,\kappa}} \in \mathcal{S}_{\mathcal{N}_i^{E,\kappa}} = \prod_{j\in\mathcal{N}_i^{E,\kappa}} \mathcal{S}_j$ and $a_{\mathcal{N}_i^{E,\kappa}} \in \mathcal{A}_{\mathcal{N}_i^{E,\kappa}} = \prod_{j\in\mathcal{N}_i^{E,\kappa}} \mathcal{A}_j$ denote the aggregated states and actions of agent i's κ-hop neighbors.

State transition probability function: $\mathcal{P}_i(s_i' | s_{\mathcal{N}_i^E}, a_i)$ specifies that agent i's next state is determined by its neighbors' states and its own action, with the global transition $\boldsymbol{\mathcal{P}}(\boldsymbol{s}' | \boldsymbol{s}, \boldsymbol{a}) = \prod_{i=1}^N \mathcal{P}_i(s_i' | s_{\mathcal{N}_i^E}, a_i)$.

Reward functions: $f_i(s_i, a_i)$ and $g_i(s_i, a_i)$ denote the objective and constraint reward functions of agent i, respectively.

Coupled policy: $\pi_i(\cdot | s_{\mathcal{N}_i^{E,\kappa_p}}, \theta_i, \theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})$ represents the local coupled policy of agent i, where $\kappa_p \geq 1$ is the coupling distance, $\theta_i \in \Theta_i \subseteq \mathbb{R}^{d_{\theta_i}}$ is the local policy parameter, and $\theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}}$ are the policy parameters of agent i's κ_p-hop neighbors. The joint policy is $\boldsymbol{\pi_\theta}(\boldsymbol{a}|\boldsymbol{s}) = \prod_{i=1}^N \pi_i(a_i | s_{\mathcal{N}_i^{E,\kappa_p}}, \theta_i, \theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})$ with joint parameter $\boldsymbol{\theta} = (\theta_1^\top, \cdots, \theta_N^\top)^\top \in \boldsymbol{\Theta} = \prod_{i=1}^N \Theta_i$. This coupled policy framework is naturally suited for environments with coupled state transition dynamics, such as intelligent transportation systems [20]–[22], where the next state of each agent is jointly determined by the current states of its neighbors. It can also be readily implemented as a neural network, where agent i's

policy takes $s_{\mathcal{N}_i^{E,\kappa_p}}$ and $\theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}}$ as inputs while each agent optimizes solely its own parameter θ_i.

Remark 1: The coupled policy $\pi_i(\cdot|s_{\mathcal{N}_i^{E,\kappa_p}}, \theta_i, \theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})$ is motivated by empirical findings in cooperative multi-agent applications, where coupled policies conditioning on both state and policy information from neighbors consistently outperform independent policies [20]–[22]. Although coupled policies have demonstrated empirical success, their theoretical foundations in distributed constrained learning remain largely unexamined. This work fills this gap by providing a rigorous convergence analysis under the coupled policy setting.

Initial state distribution: $\boldsymbol{\rho}$ is the distribution of the initial state $\boldsymbol{s}_0$.

Discount factor: $\gamma \in (0,1)$ is the discount factor.

In the CMARL problem, define the local objective function and constraint function of agent i under joint policy $\boldsymbol{\pi_\theta}$ as

$$F_i(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{s}\sim\boldsymbol{\rho}}\Big[\sum_{t=0}^{\infty}\gamma^t f_i(s_{i,t},a_{i,t})\Big|\boldsymbol{s}_0=\boldsymbol{s},\boldsymbol{a}_t\sim\boldsymbol{\pi_\theta}(\cdot|\boldsymbol{s}_t)\Big] \quad (2)$$

and

$$G_i(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{s}\sim\boldsymbol{\rho}}\Big[\sum_{t=0}^{\infty}\gamma^t g_i(s_{i,t},a_{i,t})\Big|\boldsymbol{s}_0=\boldsymbol{s},\boldsymbol{a}_t\sim\boldsymbol{\pi_\theta}(\cdot|\boldsymbol{s}_t)\Big], \quad (3)$$

where $\boldsymbol{s}_t = (s_{1,t},\cdots,s_{N,t})$ and $\boldsymbol{a}_t = (a_{1,t},\cdots,a_{N,t})$ represent the global state and global action at time t, respectively. The optimization objective of agents is to find a joint policy parameter $\boldsymbol{\theta}$ that maximizes the discounted average cumulative rewards while satisfying safety constraints:

$$\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} F(\boldsymbol{\theta}) \triangleq \frac{1}{N}\sum_{i=1}^{N} F_i(\boldsymbol{\theta}), \ \text{ s.t. } G_i(\boldsymbol{\theta}) \geq c_i, \quad (4)$$

where $c_i \in \mathbb{R}$ denotes the constraint threshold value for $G_i(\boldsymbol{\theta})$ of agent i [1].

B. Approximate first-order stationary point

The Lagrangian function associated with (4) is

$$\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu}) = F(\boldsymbol{\theta}) + \frac{1}{N}\sum_{i=1}^{N}\mu_i\big(G_i(\boldsymbol{\theta})-c_i\big), \quad (5)$$

where $\boldsymbol{\mu} = (\mu_1,\cdots,\mu_N)^\top \in \mathbb{R}_+^N$ denotes the vector of Lagrange multipliers. Accordingly, the Lagrangian formulation of (4) is expressed as

$$\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta}}\min_{\boldsymbol{\mu}\geq\mathbf{0}_N} \mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu}). \quad (6)$$

For the subsequent theoretical development, we introduce the following assumption and definition.

Assumption 1: In CMARL problem (4), there exists a first-order stationary point (FOSP) $(\boldsymbol{\theta}^\star,\boldsymbol{\mu}^\star)$ and a positive constant $\mu_{max} > 0$, such that $\mu_i^\star \leq \mu_{max}$ holds for all $i \in \mathcal{N}$.

Assumption 1 adheres to the standard practice in constrained optimization by assuming that the feasible set of the dual variables is bounded, which is consistent with [3], [16].

Since $F(\boldsymbol{\theta})$ and $\{G_i(\boldsymbol{\theta})\}_{i\in\mathcal{N}}$ are generally nonconvex in $\boldsymbol{\theta}$, computing the global optimum of (4) is NP-hard. In this work, we instead seek an approximate FOSP of problem (4). Under Assumption 1, let $\boldsymbol{\mathcal{U}} = \prod_{i=1}^N \mathcal{U}_i$ as the feasible set of the dual variables, where $\mathcal{U}_i = [0,\mu_{max}]$ for each agent $i \in \mathcal{N}$. The ϵ-FOSP is defined as follows.

Definition 1: (ϵ-FOSP) A solution $(\boldsymbol{\theta},\boldsymbol{\mu}) \in \boldsymbol{\Theta}\times\boldsymbol{\mathcal{U}}$ is said to be an ϵ-FOSP, if

$$\mathcal{E}(\boldsymbol{\theta},\boldsymbol{\mu}) = \mathcal{X}(\boldsymbol{\theta},\boldsymbol{\mu})^2 + \mathcal{Y}(\boldsymbol{\theta},\boldsymbol{\mu})^2 \leq \epsilon, \quad (7)$$

where $\mathcal{X}(\boldsymbol{\theta},\boldsymbol{\mu})$ and $\mathcal{Y}(\boldsymbol{\theta},\boldsymbol{\mu})$ are respectively defined as

$$\mathcal{X}(\boldsymbol{\theta},\boldsymbol{\mu}) = -\min_{\boldsymbol{\mu}'\in\boldsymbol{\mathcal{U}},\|\boldsymbol{\mu}'-\boldsymbol{\mu}\|\leq 1}\langle\nabla_{\boldsymbol{\mu}}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu}),\boldsymbol{\mu}'-\boldsymbol{\mu}\rangle \quad (8)$$

and

$$\mathcal{Y}(\boldsymbol{\theta},\boldsymbol{\mu}) = \max_{\boldsymbol{\theta}'\in\boldsymbol{\Theta},\|\boldsymbol{\theta}'-\boldsymbol{\theta}\|\leq 1}\langle\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu}),\boldsymbol{\theta}'-\boldsymbol{\theta}\rangle. \quad (9)$$

Since $\mathcal{X}(\boldsymbol{\theta},\boldsymbol{\mu})$ in (8) and $\mathcal{Y}(\boldsymbol{\theta},\boldsymbol{\mu})$ in (9) are defined based on the first-order optimality conditions [23], if there exists $(\boldsymbol{\theta}^*,\boldsymbol{\mu}^*) \in \boldsymbol{\Theta}\times\boldsymbol{\mathcal{U}}$ such that $\mathcal{E}(\boldsymbol{\theta}^*,\boldsymbol{\mu}^*) = 0$, then $(\boldsymbol{\theta}^*,\boldsymbol{\mu}^*)$ constitutes a FOSP of problem (4).

C. Coupled policy gradient

For the CMARL problem in (4), given any joint policy $\boldsymbol{\pi_\theta}$ and Lagrangian multiplier $\boldsymbol{\mu}$, we define the global and local Lagrangian Q-functions as

$$\begin{aligned} Q^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}) =& \mathbb{E}_{\boldsymbol{\pi_\theta}}\Big[\frac{1}{N}\sum_{t=0}^{\infty}\sum_{i=1}^{N}\gamma^t\big(f_i(s_{i,t},a_{i,t}) \\ &+ \mu_i g_i(s_{i,t},a_{i,t})\big)\Big|\boldsymbol{s}_0=\boldsymbol{s},\boldsymbol{a}_0=\boldsymbol{a}\Big] \end{aligned} \quad (10)$$

and

$$\begin{aligned} Q_i^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}) =& \mathbb{E}_{\boldsymbol{\pi_\theta}}\Big[\sum_{t=0}^{\infty}\gamma^t\big(f_i(s_{i,t},a_{i,t})+\mu_i g_i(s_{i,t},a_{i,t})\big)\Big| \\ &\boldsymbol{s}_0=\boldsymbol{s},\boldsymbol{a}_0=\boldsymbol{a}\Big]. \end{aligned} \quad (11)$$

Based on the above definitions, the relationship between the global and local Lagrangian Q-functions satisfies

$$Q^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}) = \frac{1}{N}\sum_{i=1}^{N} Q_i^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}). \quad (12)$$

We further define $Q_{g,i}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a})$ as the local constraint Q-function for agent i:

$$Q_{g,i}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a}) = \mathbb{E}_{\boldsymbol{\pi_\theta}}\Big[\sum_{t=0}^{\infty}\gamma^t g_i(s_{i,t},a_{i,t})\Big|\boldsymbol{s}_0=\boldsymbol{s},\boldsymbol{a}_0=\boldsymbol{a}\Big]. \quad (13)$$

Denote $d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s})$ as the discounted state visitation distribution of $\boldsymbol{s} \in \boldsymbol{\mathcal{S}}$ generated by the joint policy $\boldsymbol{\pi_\theta}$ and satisfy

$$d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s}) = (1-\gamma)\sum_{t=0}^{\infty}\gamma^t \mathrm{Pr}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s}_t=\boldsymbol{s}|\boldsymbol{s}_0\sim\boldsymbol{\rho}), \quad (14)$$

where $\mathrm{Pr}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s}_t=\boldsymbol{s}|\boldsymbol{s}_0\sim\boldsymbol{\rho})$ is the probability of occurrence of $\boldsymbol{s}_t=\boldsymbol{s}$ at time t under joint policy $\boldsymbol{\pi_\theta}$ and initial state distribution $\boldsymbol{\rho}$.

Based on the above definitions, the coupled policy gradients of $\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ with respect to θ_i and μ_i are given in the following lemma.

[1]Although the unconstrained cooperative objective admits a MPG formulation, our constrained formulation (4) is analyzed through the Lagrangian, and the solution concept is an ϵ-FOSP rather than a Nash equilibrium.

Lemma 1: In CMARL problem (4), for any joint policy $\boldsymbol{\pi_\theta}$, the gradients of $\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ with respect to θ_i and μ_i are respectively represented as

$$\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu}) = \frac{1}{N}\big(G_i(\boldsymbol{\theta}) - c_i\big) \tag{15}$$

and

$$\begin{aligned}\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu}) =& \frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[Q^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}) \\ &\times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big].\end{aligned} \tag{16}$$

Lemma 1 provides the explicit expressions for the gradients of both the Lagrange multiplier and the coupled policy, with the corresponding proof detailed in Section IX-A.

IV. DSPD ALGORITHM

In this section, we introduce the approximated coupled policy gradient and propose the DSPD algorithm to address the CMARL problem under coupled policies.

A. Approximated policy gradient

To reduce the reliance on global state-action information, we adopt a truncation-based approach [19] and design a novel neighbors' averaged Lagrangian Q-function for each agent i:

$$\begin{aligned}\widehat{Q_i^{\boldsymbol{\pi_\theta}}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}) =& \mathbb{E}_{\boldsymbol{\pi_\theta}}\Big[\frac{1}{N}\sum_{t=0}^{\infty}\gamma^t \sum_{j\in\mathcal{N}_i^{E,\kappa+2\kappa_p}}\big(f_j(s_{j,t},a_{j,t}) \\ &+ \mu_j g_j(s_{j,t},a_{j,t})\big)\Big|\boldsymbol{s}_0=\boldsymbol{s},\boldsymbol{a}_0=\boldsymbol{a}\Big].\end{aligned} \tag{17}$$

The $(\kappa+2\kappa_p)$-hop neighborhood in (17) arises naturally from two components of the coupled policy gradient (16): the log-policy gradients of κ_p-hop neighbors require state-action pairs from their own κ_p-hop neighborhoods (i.e., the $2\kappa_p$-hop neighborhood of agent i), and inspired by the κ-distance truncation in [19], an additional κ-hop range is introduced to approximate the global Lagrangian Q-function, yielding the total range of $\kappa+2\kappa_p$ hops. Based on (17), the approximated policy gradient for agent i is

$$\begin{aligned}g_{app,i}^{\boldsymbol{\pi_\theta},\boldsymbol{\mu}} =& \frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[\widehat{Q_i^{\boldsymbol{\pi_\theta}}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}) \\ &\times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big],\end{aligned} \tag{18}$$

which replaces the global Lagrangian Q-function in (16) with the neighbors' averaged Lagrangian Q-function, offering an approximation to $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$. Since $\widehat{Q_i^{\boldsymbol{\pi_\theta}}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})$ can be estimated from reward signals within the $(\kappa+2\kappa_p)$-hop neighborhood and $g_{app,i}^{\boldsymbol{\pi_\theta},\boldsymbol{\mu}}$ from state-action pairs within the $2\kappa_p$-hop neighborhood, this provides a theoretical basis for fully distributed implementation.

B. Estimation of agent-specific parameters

In the m-th iteration, let $\boldsymbol{\theta}_m$ and $\boldsymbol{\mu}_m$ denote the joint policy parameter and Lagrangian multiplier. To reduce the exposure of sensitive learned parameters, each agent exchanges true values only with immediate neighbors over the time-varying learning networks $\{\mathcal{G}_m^L\}_{m\geq 1}$, and maintains local estimates of other agents' parameters. Specifically, we define $\hat{\boldsymbol{\theta}}_m^i = \big((\hat{\theta}_{1,m}^i)^\top,\cdots,(\hat{\theta}_{N,m}^i)^\top\big)^\top$ and $\hat{\boldsymbol{\mu}}_m^i = \big((\hat{\mu}_{1,m}^i)^\top,\cdots,(\hat{\mu}_{N,m}^i)^\top\big)^\top$ as agent i's estimates of other agents' policy parameters and Lagrange multipliers, and $\breve{\boldsymbol{\theta}}_m^i = \big((\breve{\theta}_{1,m}^i)^\top,\cdots,(\theta\mu_{N,m}^i)^\top\big)^\top$, $\breve{\boldsymbol{\mu}}_m^i = \big((\breve{\mu}_{1,m}^i)^\top,\cdots,(\breve{\mu}_{N,m}^i)^\top\big)^\top$ as the corresponding intermediate variables in the learning process.

Assumption 2: During the learning process, each agent i receives state-action pairs $\{(s_j,a_j)\}_{j\in\mathcal{N}_i^{E,2\kappa_p}}$ and reward signals $\{f_j,g_j\}_{j\in\mathcal{N}_i^{E,\kappa+2\kappa_p}}$ through $\mathcal{G}^E$, and exchanges intermediate variables $\{\hat{\boldsymbol{\theta}}_m^i,\breve{\boldsymbol{\mu}}_m^i\}$ and true values $\{\theta_{i,m},\mu_{i,m}\}$ with direct neighboring agents over $\{\mathcal{G}_m^L(\mathcal{N},\mathcal{E}_m^L)\}_{m\geq 1}$.

Assumption 2 introduces a dedicated learning network $\{\mathcal{G}_m^L\}_{m\geq 1}$, separate from the environmental network $\mathcal{G}^E$, motivated by practical systems such as networked control systems [24] and digital twin-enabled systems [25], where the communication infrastructure for parameter exchange naturally differs from that for environmental interactions. This design limits the exposure of each agent's decision-making variables to a minimal set of neighbors, thereby reducing vulnerability to potential network attacks compared to frameworks requiring global parameter broadcast [3].

C. DSPD algorithm designing

Under Assumption 2, agents receive states from theirs κ_p-hop environmental neighbors and execute joint policy $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m} = \prod_{i=1}^N \pi_i(a_i|s_{\mathcal{N}_i^{E,\kappa_p}},\hat{\theta}_{i,m}^i,\hat{\theta}^i_{\mathcal{N}_{i,-i}^{E,\kappa_p},m})$. The DSPD algorithm proceeds in three steps.

Step 1. Parameter estimation: In the m-th iteration, the estimates $\hat{\mu}_{j,m}^i$ and $\hat{\theta}_{j,m}^i$ are computed via the push-sum protocol [17]:

$$\begin{cases} p_{i,m+1} = \displaystyle\sum_{l\in\mathcal{N}_{i,m}^L} w_{il,m}^L p_{l,m} & \text{(19a)}\\ \hat{\mu}_{j,m}^i = \dfrac{1}{p_{i,m+1}}\displaystyle\sum_{l\in\mathcal{N}_{i,m}^L} w_{il,m}^L \breve{\mu}_{j,m}^l & \text{(19b)}\\ \hat{\theta}_{j,m}^i = \dfrac{1}{p_{i,m+1}}\displaystyle\sum_{l\in\mathcal{N}_{i,m}^L} w_{il,m}^L \breve{\theta}_{j,m}^l, & \text{(19c)}\end{cases}$$

with $\breve{\mu}_{j,1}^i = 0$, $\breve{\theta}_{j,1}^i = \mathbf{0}_{d_{\theta_j}}$, and $p_{i,1}=1$ for all $i,j\in\mathcal{N}$.

Step 2. Lagrangian multiplier update: Each agent i estimates $\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_m)$ using K_μ batches of samples. For each batch $k\in\{1,\cdots,K_\mu\}$, agents execute $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$ to generate a sample trajectory $\{\boldsymbol{s},\boldsymbol{a}\}_{0:T_1(k)}$ of length $T_1(k)\sim\mathrm{Geom}(1-\gamma^{1/2})$, and each agent i computes the gradient estimate:

$$\hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k) = \frac{1}{N}\Big(\sum_{t=0}^{T_1(k)}\gamma^{t/2}g_{i,t} - c_i\Big). \tag{20}$$

Based on (20), we define the averaged estimate $\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}$ as

$$\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m} = \frac{1}{K_\mu}\sum_{k=1}^{K_\mu} \hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k) \tag{21}$$

and update the Lagrangian multiplier $\mu_{i,m+1}$ according to

$$\mu_{i,m+1} = P_{\mathcal{U}_i}(\mu_{i,m} - \eta_{\mu,m}\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}), \tag{22}$$

where $\eta_{\mu,m}$ is the learning rate of Lagrangian multiplier.

The intermediate variable of the Lagrangian multipliers $\breve{\mu}_{j,m+1}^i$ is then revised as

$$\breve{\mu}_{j,m+1}^i = \begin{cases} \sum\limits_{l\in\mathcal{N}_{i,m}^L} w_{il,m}^L \breve{\mu}_{j,m}^l, \text{ if } j \notin \mathcal{N}_{i,m}^L \\ \sum\limits_{l\in\mathcal{N}_{i,m}^L} w_{il,m}^L \breve{\mu}_{j,m}^l + w_{ij,m}^L N(\mu_{j,m+1}-\mu_{j,m}), \text{o/w}, \end{cases} \tag{23}$$

which ensures $(1/N)\sum_{i=1}^N \breve{\mu}_{j,m}^i = \mu_{j,m}$, guaranteeing average consensus convergence of $\hat{\mu}_{j,m}^i$ towards $\mu_{j,m}$.

Step 3. Policy parameter update: After Step 2, each agent i first updates $\hat{\boldsymbol{\mu}}_{m+1}^i$ via (19b). Analogous to Step 2, each agent i estimates $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_{m+1})$ using K_θ batches of samples. For each batch $k \in \{1,\cdots,K_\theta\}$, agents execute $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$ to generate two sample trajectories $\{\boldsymbol{s},\boldsymbol{a}\}_{0:T_2(k)}$ and $\{\boldsymbol{s},\boldsymbol{a}\}_{T_2(k):T_2(k)+T_3(k)}$, where $T_2(k) \sim \text{Geom}(1-\gamma)$ and $T_3(k) \sim \text{Geom}(1-\gamma^{1/2})$, and each agent i computes the approximated policy gradient estimate:

$$\hat{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}(k) = \frac{1}{1-\gamma}\hat{Q}_{i,T_2(k)}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}} \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i}\log\pi_j(a_{j,T_2(k)}| s_{\mathcal{N}_j^{E,\kappa_p},T_2(k)}, \hat{\theta}_{j,m}^i, \hat{\theta}_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}^i), \tag{24}$$

where $\hat{Q}_{i,T_2(k)}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ is a sample-based estimate of the neighbors' averaged Lagrangian Q-function $\widehat{Q_i^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}}(\boldsymbol{s}_{T_2(k)}, \boldsymbol{a}_{T_2(k)}; \hat{\boldsymbol{\mu}}_{m+1}^i)$ in (17), computed from the second trajectory $\{\boldsymbol{s},\boldsymbol{a}\}_{T_2(k):T_2(k)+T_3(k)}$:

$$\hat{Q}_{i,T_2(k)}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}} = \frac{1}{N}\sum_{t=0}^{T_3(k)} \gamma^{t/2} \sum_{j\in\mathcal{N}_i^{E,\kappa+2\kappa_p}} \big(f_{j,T_2(k)+t} + \hat{\mu}_{j,m+1}^i g_{j,T_2(k)+t}\big). \tag{25}$$

Define the averaged policy gradient estimate for agent i as

$$\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}} = \frac{1}{K_\theta}\sum_{k=1}^{K_\theta} \hat{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}(k), \tag{26}$$

the policy parameter is updated according to

$$\theta_{i,m+1} = P_{\Theta_i}(\theta_{i,m} + \eta_{\theta,m}\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}), \tag{27}$$

where $\eta_{\theta,m}$ is the learning rate of policy parameter.

Remark 2: The use of $\hat{\boldsymbol{\mu}}_{m+1}$ instead of $\hat{\boldsymbol{\mu}}_m$ in $\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ in (26) is advantageous, as it facilitates the theoretical analysis of Theorem 3, where the ϵ-policy stationary convergence is rigorously established.

Similar to (23), the intermediate variable for policy parameters is updated as

$$\breve{\theta}_{j,m+1}^i = \begin{cases} \sum\limits_{l\in\mathcal{N}_{i,m}^L} w_{il,m}^L \breve{\theta}_{j,m}^l, \text{ if } j \notin \mathcal{N}_{i,m}^L \\ \sum\limits_{l\in\mathcal{N}_{i,m}^L} w_{il,m}^L \breve{\theta}_{j,m}^l + w_{ij,m}^L N(\theta_{j,m+1}-\theta_{j,m}), \text{o/w}, \end{cases} \tag{28}$$

which ensures $(1/N)\sum_{i=1}^N \breve{\theta}_{j,m}^i = \theta_{j,m}$ and average consensus convergence of $\hat{\theta}_{j,m}^i$ to $\theta_{j,m}$.

To integrate Steps 1-3, the pseudocode of the DSPD algorithm is proposed in Algorithm 1.

As illustrated in Algorithm 1, the DSPD algorithm operates in a fully distributed manner involving two types of information: environmental data (state-action pairs and rewards) and sensitive learned information (policy parameters and Lagrange multipliers). Specifically, each agent i receives state-action pairs and rewards through $\mathcal{G}^E$ (refer to Line 15), and exchanges estimated and true parameter values with direct neighbors through $\mathcal{G}_m^L$ (refer to Lines 4, 11, and 18). The information exchange structure is summarized in Table I.

TABLE I
SUMMARY OF INFORMATION EXCHANGE OF AGENT i IN THE DSPD ALGORITHM.

Information	Range	Network
State-action pairs	$2\kappa_p$-hop	$\mathcal{G}^E$
Reward/constraint signals	$(\kappa+2\kappa_p)$-hop	$\mathcal{G}^E$
True parameters $\{\theta_{i,m}, \mu_{i,m}\}$	1-hop	$\mathcal{G}_m^L$
Intermediate variables $\{\breve{\theta}_{j,m}^i, \breve{\mu}_{j,m}^i\}$	1-hop	$\mathcal{G}_m^L$

Remark 3: Although agents execute the coupled policy using estimated parameters $\{\hat{\theta}_{j,m}^i\}_{j\in\mathcal{N}}$, the coupling is genuine in two aspects: (i) the state information $s_{\mathcal{N}_i^{E,\kappa_p}}$ is directly accessible through $\mathcal{G}^E$; and (ii) as established in Theorem 2, $\lim_{m\to\infty}\hat{\theta}_{j,m}^i = \theta_{j,m}$, ensuring the executed policy asymptotically converges to the intended coupled policy.

V. CONVERGENCE RESULTS

Before proceeding with the analysis, we introduce a set of standard assumptions [3], [16], [19], [26], [27] as follows.

Assumption 3: In CMARL problem (4), there exists $R_f, R_g > 0$ such that $|f_i(s_i,a_i)| \le R_f$ and $|g_i(s_i,a_i)| \le R_g$ for all $i \in \mathcal{N}$ and $(s_i,a_i) \in \mathcal{S}_i \times \mathcal{A}_i$.

Assumption 4: For any joint policy parameter $\boldsymbol{\theta}$, $d_{\boldsymbol{\rho}}^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}(\boldsymbol{s})$ and $\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}(\boldsymbol{s},\boldsymbol{a})$ satisfy $\inf_{\boldsymbol{\theta}}\min_{\boldsymbol{s}\in\boldsymbol{\mathcal{S}}} d_{\boldsymbol{\rho}}^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}(\boldsymbol{s}) > 0$ and $\inf_{\boldsymbol{\theta}}\min_{(\boldsymbol{s},\boldsymbol{a})\in\boldsymbol{\mathcal{S}}\times\boldsymbol{\mathcal{A}}} \xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}(\boldsymbol{s},\boldsymbol{a}) > 0$.

Assumption 4 indicates that the coupled policies explore all the states and actions with some positive probability, which is a common assumption in the related literature [26], [27].

Assumption 5: For any joint policy $\boldsymbol{\pi}_{\boldsymbol{\theta}}$ and agent $i \in \mathcal{N}$, the local policy is differentiable with respect to $\{\theta_j\}_{j\in\mathcal{N}_i^{E,\kappa_p}}$. For any $j \in \mathcal{N}_i^{E,\kappa_p}$, the gradient $\nabla_{\theta_j}\log\pi_i(a_i|s_{\mathcal{N}_i^{E,\kappa_p}}, \theta_i, \theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})$ exists, satisfies $\|\nabla_{\theta_j}\log\pi_i(a_i|s_{\mathcal{N}_i^{E,\kappa_p}}, \theta_i, \theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})\| \le B$ for some $B > 0$, and is L-Lipschitz continuous with respect to θ_j.

Assumption 5 implies that the gradients of the log of the policy functions have bounded norms and exhibit Lipschitz continuity. Specifically, this assumption is satisfied by common policy parameterizations such as softmax or log-linear policies.

Algorithm 1: The DSPD Algorithm

Require: $M, K_\mu, K_\theta \in \mathbb{N}^+$, learning rates $\eta_{\mu,m}$ and $\eta_{\theta,m}$;
Initialize: $\mu_{j,1} = \breve{\mu}^i_{j,1} = 0$, $\theta_{j,1} = \breve{\theta}^i_{j,1} = \mathbf{0}_{d_{\theta_i}}$, $\forall i, j \in \mathcal{N}$;
for $m = 1, \cdots, M$ **do**
  Compute $\{\hat{\mu}^i_{j,m}\}_{j\in\mathcal{N}}$ and $\{\hat{\theta}^i_{j,m}\}_{j\in\mathcal{N}}$ via (19);
  /* Distributed update of Lagrangian multipliers */
  for $k = 1, \cdots, K_\mu$ **do**
    Draw $\boldsymbol{s}_0 \sim \boldsymbol{\rho}$, $T_1(k) \sim \text{Geom}(1-\gamma^{1/2})$;
    Agents execute $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$ to generate $\{\boldsymbol{s}, \boldsymbol{a}\}_{0:T_1(k)}$;
    Each agent i computes $\hat{h}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_m}_{app,i}(k)$ via (20);
  end
  Each agent i computes $\bar{h}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_m}_{app,i}$ via (21) and updates $\mu_{i,m+1}$ via (22);
  Each agent i updates $\{\breve{\mu}^i_{j,m+1}\}_{j\in\mathcal{N}}$ via (23) and $\{\hat{\mu}^i_{j,m+1}\}_{j\in\mathcal{N}}$ via (19b);
  /* Distributed update of policy parameters */
  for $k = 1, \cdots, K_\theta$ **do**
    Draw $\boldsymbol{s}_0 \sim \boldsymbol{\rho}$, $T_2(k) \sim \text{Geom}(1-\gamma)$, $T_3(k) \sim \text{Geom}(1-\gamma^{1/2})$;
    Agents execute $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$ to generate $\{\boldsymbol{s}, \boldsymbol{a}\}_{0:T_2(k)}$ and $\{\boldsymbol{s}, \boldsymbol{a}\}_{T_2(k):T_2(k)+T_3(k)}$;
    Each agent i collects $(s_{\mathcal{N}_i^{E,2\kappa_p}, T_2(k)}, a_{\mathcal{N}_i^{E,2\kappa_p}, T_2(k)})$ and $\{f_{\mathcal{N}_i^{E,\kappa+2\kappa_p}, T_2(k)+t}, g_{\mathcal{N}_i^{E,\kappa+2\kappa_p}, T_2(k)+t}\}_{t=0}^{T_3(k)}$ to compute $\hat{g}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_{m+1}}_{app,i}(k)$ via (24);
  end
  Each agent i computes $\bar{g}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_{m+1}}_{app,i}$ via (26) and updates $\theta_{i,m+1}$ via (27);
  Each agent i updates $\{\breve{\theta}^i_{j,m+1}\}_{j\in\mathcal{N}}$ via (28);
end
Output: $\{\boldsymbol{\theta}_m\}_{m=1}^{M+1}$ and $\{\boldsymbol{\mu}_m\}_{m=1}^{M+1}$.

Assumption 6: The time-varying learning networks $\{\mathcal{G}_m^L\}_{m\geq 1}$ are uniformly strongly connected.

Assumption 7: In Algorithm 1, the learning rates $\eta_{\theta,m}$ and $\eta_{\mu,m}$ are set such that $\eta_{\mu,m} \sim \mathcal{O}(\frac{1}{m})$ and $\eta_{\theta,m} \sim \mathcal{O}(\frac{1}{\sqrt{m}})$.

Assumptions 6-7 are standard conditions on time-varying networks and learning rates essential for guaranteeing convergence.

A. Main results

Three main theoretical contributions of the proposed DSPD algorithm are established as follows. Theorem 1 bounds the approximation error of the approximated coupled policy gradient, involving a novel joint function of both κ and κ_p. Theorem 2 establishes the convergence of the distributed parameter estimates via a carefully ordered interleaved analysis. Theorem 3 establishes the overall ϵ-policy stationary convergence of the proposed algorithm.

Theorem 1 (Approximated policy gradient error): Suppose Assumptions 1 and 3-4 hold. For any joint policy $\boldsymbol{\pi}_{\boldsymbol{\theta}}$, Lagrangian multiplier $\boldsymbol{\mu}$, and agent $i \in \mathcal{N}$, the error between the approximated policy gradient $g^{\boldsymbol{\pi}_{\boldsymbol{\theta}}, \boldsymbol{\mu}}_{app,i}$ and the exact gradient $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\mu})$ satisfies

$$\begin{aligned}&\|g^{\boldsymbol{\pi}_{\boldsymbol{\theta}}, \boldsymbol{\mu}}_{app,i} - \nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\mu})\| \\ \leq &\frac{2(R_f + \mu_{max}R_g)BN}{(1-\gamma)^2}\gamma^{h(\kappa,\kappa_p)+1},\end{aligned} \tag{29}$$

where $h(\kappa, \kappa_p)$ is defined as

$$h(\kappa, \kappa_p) = \begin{cases} \frac{\kappa+1}{\kappa_p} - 1, & \text{if } \text{mod}(\kappa+1, \kappa_p) = 0 \\ \lfloor \frac{\kappa+1}{\kappa_p} \rfloor, & \text{otherwise.} \end{cases} \tag{30}$$

The detailed analysis of Theorem 1 is provided in Section VIII-A. Theorem 1 establishes an upper bound on the approximation error of the coupled policy gradient. Compared to existing works [3], [19] where the approximation error depends only on κ, the error bound here involves the joint function $h(\kappa, \kappa_p)$ of both κ and κ_p, reflecting the additional approximation complexity introduced by the coupled policy structure.

For the constraints $\{c_i\}_{i\in\mathcal{N}}$ in problem (4), we define $c_{max} = \max_{i\in\mathcal{N}} |c_i|$ and have the following theorem.

Theorem 2 (Convergence of parameter estimations): Suppose Assumptions 1-7 hold. In Algorithm 1, for any $i, j \in \mathcal{N}$, there exists $\tilde{\mu}_{max} > \mu_{max}$, $L_1^\mu = \frac{R_g + c_{max}}{(1-\gamma^{1/2})N}$, and $L_1^\theta = \frac{BN(R_f + \tilde{\mu}_{max}R_g)}{(1-\gamma)(1-\gamma^{1/2})}$, such that

(i) $\|\hat{h}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_m}_{app,i}(k)\| \leq L_1^\mu$ for all $k \in \{1, \cdots, K_\mu\}$ and $\lim_{m\to\infty} \hat{\mu}^i_{j,m} = \mu_{j,m}$ with rate $\mathcal{O}(\frac{1}{m})$;

(ii) $\|\hat{g}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_{m+1}}_{app,i}(k)\| \leq L_1^\theta$ for all $k \in \{1, \cdots, K_\theta\}$ and $\lim_{m\to\infty} \hat{\theta}^i_{j,m} = \theta_{j,m}$ with rate $\mathcal{O}(\frac{1}{\sqrt{m}})$.

The proof of Theorem 2 can be found in Section VIII-B. Theorem 2 demonstrates that all parameter estimates converge to their true values, ensuring the accuracy of the policies executed by the agents and providing reliable gradient information for the updates in (22) and (27). The convergence of both estimates is established through a carefully ordered interleaved analysis that resolves the circular dependency between the policy parameter and Lagrange multiplier estimation processes.

Theorem 3 (ϵ-policy stationary convergence): Suppose Assumptions 1-7 hold. For every $\epsilon > 0$ and $\delta > 0$, let $\{(\boldsymbol{\theta}_m, \boldsymbol{\mu}_m)\}_{m=1}^M$ be the sequence generated by Algorithm 1 with $M = \mathcal{O}(1/\epsilon^2)$, $\eta_{\theta,m} = 1/(2\sqrt{m} + L_{\theta\theta})$, $\eta_{\mu,m} = 1/2m$, and $K_\theta = K_\mu = \log(2/\delta)/2\epsilon^2$, where $L_{\theta\theta} = LN^{\frac{3}{2}}(R_f + \tilde{\mu}_{max}R_g)/(1-\gamma)^2 + (1+\gamma)B^2N^3(R_f + \tilde{\mu}_{max}R_g)/(1-\gamma)^3$ is Lipschitz constant. Then, with probability $1-\delta$, there exists $m^* \in \{1, \cdots, M-1\}$ such that

$$\left[\mathcal{Y}(\boldsymbol{\theta}_{m^*}, \boldsymbol{\mu}_{m^*+1})\right]^2 = \mathcal{O}(\epsilon) + \mathcal{O}\big(\epsilon_0(\kappa, \kappa_p)^2\big), \tag{31}$$

where $\epsilon_0(\kappa, \kappa_p) = 2(R_f + \tilde{\mu}_{max}R_g)BN^{\frac{3}{2}}\gamma^{h(\kappa,\kappa_p)+1}/(1-\gamma)^2$.

The detailed analysis of Theorem 3 is provided in Section VIII-C. Theorem 3 establishes that, with high probability, our Algorithm 1 achieves an ϵ-policy stationary convergence with an approximation error of $\mathcal{O}\big(\epsilon_0(\kappa, \kappa_p)^2\big)$. This error term reveals the following trade-offs: (i) increasing κ leads to a larger $h(\kappa, \kappa_p)$, reducing the approximation error at the cost of increased communication overhead; (ii) decreasing κ_p also leads to a larger $h(\kappa, \kappa_p)$, reducing the approximation error but limiting the ability of agents to capture inter-agent dependencies. Therefore, κ and κ_p should be chosen to balance communication overhead and approximation accuracy in practice. We remark that Theorem 3 only establishes

stationary with respect to the policy parameter $\boldsymbol{\theta}$. In general, convergence of a primal-dual algorithm to a stationary point of the Lagrangian function does not imply that the resulting policy is feasible. Establishing a rigorous convergence rate for constraint violation in our distributed, coupled-policy setting therefore remains an important direction for future research. In Section VI, we validate constraint satisfaction empirically.

VI. Simulation

In this section, we study the empirical performance of the proposed Algorithm 1 on a wireless access-control network with safety constraints [3].

A. Wireless access-control network environment

We consider a cooperative task involving $\mathcal{N} = \{1, \cdots, 25\}$ transmitting agents within a 5×5 wireless access-control network, which comprises 16 access points $Y = \{1, \cdots, 16\}$, as illustrated in Fig. 1.

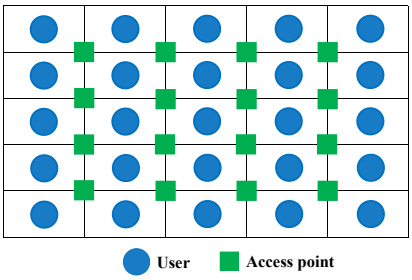


Fig. 1. Wireless access-control network.

Each agent $i \in \mathcal{N}$ is connected to a set $Y_i \subset Y$ of access points located at the corner of the block it resides in, and two agents are direct neighbors if they share a common access point, which naturally induces the environmental network $\mathcal{G}^E$. In every time slot, agent i receives a new packet by deadline $d_i^s > 0$ with probability $p_i^{new} \in (0, 1)$, and can choose to send the earliest packet in its queue to an access point $y \in Y_i$ or do nothing. Agent i receives a unit objective reward if and only if access point y does not receive transmissions from other agents and successfully processes the packet, which occurs with probability $q_y \in (0, 1)$. Otherwise, the objective reward is zero. In this environment, the state of each agent i is a d_i^s-dimensional binary vector $s_i \in \mathcal{S}_i = \{0, 1\}^{d_i^s}$, where the k-th entry takes the value 1 when agent i currently has a packet with k time steps remaining until the deadline. The action $a_i \in \mathcal{A}_i = Y_i \cup \{\text{null}\}$ of agent i specifies whether to send the earliest queued packet to an access point $y \in Y_i$ or to do nothing (i.e., $a_i = \text{null}$). The safety constraint is a cumulative transmission budget with per-step constraint reward $g_i(s_i, a_i) = -\mathbb{1}_{\{a_i \neq \text{null}\}}$. Since the local transition dynamics and reward of each agent depend on the states and actions of its neighboring agents, the problem exhibits strongly coupled state-action dynamics.

B. Parameters setting

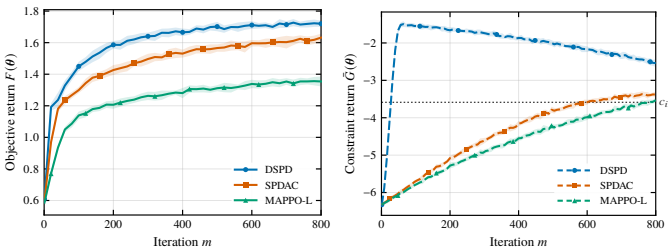
The coupled policy $\pi_i(\cdot|s_{\mathcal{N}_i^{E,\kappa_p}}, \theta_i, \theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})$ of agent i in the proposed Algorithm 1 adopts the following softmax form to capture the inter-agent dependencies:

$$\pi_i(a_i|s_{\mathcal{N}_i^{E,\kappa_p}}, \theta_i, \theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}}) = \frac{\exp\left(0.9 * \theta_{i,s_i,a_i} + \frac{0.1}{|\mathcal{N}_i^{E,\kappa_p}|-1} \sum_{j \in \mathcal{N}_{i,-i}^{\kappa_p}} \theta_{j,s_i,a_i}\right)}{\sum_{a_i'} \exp\left(0.9 * \theta_{i,s_i,a_i'} + \frac{0.1}{|\mathcal{N}_i^{E,\kappa_p}|-1} \sum_{j \in \mathcal{N}_{i,-i}^{\kappa_p}} \theta_{j,s_i,a_i'}\right)}, \tag{32}$$

where $\theta_{i,s_i,a_i} \in \mathbb{R}$ is the policy parameter of agent i corresponding to local state-action (s_i, a_i), and θ_i is the aggregation of θ_{i,s_i,a_i} across all state-action pair $(s_i, a_i) \in \mathcal{S}_i \times \mathcal{A}_i$. The parameter setting in Algorithm 1 and environment are configured as follows: $p_i^{new} = 0.5$, $q_y = 0.8$, $d_i^s = 2$, $\gamma = 0.9$, $c_i = -3.56$ for all $i \in \mathcal{N}$, $\kappa = \kappa_p = 1$, $K_\mu = 4$, $K_\theta = 1$, and $\mu_{max} = 50$. Since the transmission success of each agent is directly determined by whether its immediate neighboring agents transmit simultaneously, setting $\kappa_p = 1$ is sufficient to capture the inter-agent dependencies arising from the coupled state-action dynamics in this environment. The learning networks among agents switch between different topologies $\{0 \to 1, 2 \to 3, \cdots, 22 \to 23, 24 \to 0\}$ and $\{1 \to 2, 3 \to 4, \cdots, 23 \to 24\}$. Further implementation details and hyperparameter settings are available in our open-source code at https://github.com/EricDMW/DSPD_New.git.

C. Simulation results

To verify the effectiveness of our Algorithm 1, we compare it with the scalable primal-dual actor-critic (SPDAC) algorithm [3] that focuses on parameter-independent policies, and the MAPPO-Lagrangian (MAPPO-L) algorithm [28]. All algorithms are trained under 16 different random seeds, and the results are shown in Figs. 2-3.



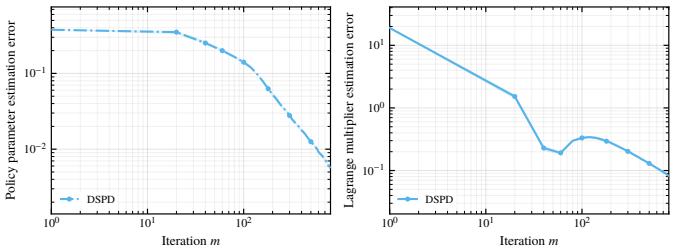
(a) Objective return (b) Average constraint return

Fig. 2. The evolution of the objective return and average constraint return of all algorithms.

Fig. 2(a) illustrates the evolution of the objective return $F(\boldsymbol{\theta})$ of all algorithms, where the solid lines represent the mean values across different random seeds and the shaded areas denote the variance. As observed, our Algorithm 1 achieves the highest objective return, clearly separated from SPDAC and MAPPO-L with non-overlapping bands. This result aligns with our hypothesis that a coupled policy framework is capable of achieving superior learning performance compared to an independent policy structure, as it more effectively captures the inter-agent dependencies.

The constraint performances of all algorithms are presented in Fig. 2(b), where all algorithms are initialized over the transmission budget and start infeasible. The proposed DSPD algorithm restores feasibility first and achieves with the largest safety margin above constraint threshold value, demonstrating its superior constraint satisfaction performance.

To further validate the accuracy of the proposed DSPD algorithm in parameter estimation, Figs. 3(a)-3(b) present the evolution of the estimation errors for policy parameters and



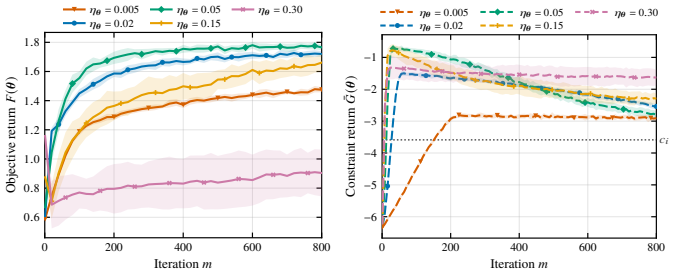
(a) $\frac{1}{N^2}\sum_{i,j=1}^{N}\|\hat{\theta}^i_{j,m}-\theta_{j,m}\|$ (b) $\frac{1}{N^2}\sum_{i,j=1}^{N}|\hat{\mu}^i_{j,m}-\mu_{j,m}|$

Fig. 3. The evolution of the estimation errors for policy parameters and Lagrange multipliers.

Lagrange multipliers throughout the learning process. As depicted, both $\frac{1}{N^2}\sum_{i,j=1}^{N}\|\hat{\theta}^i_{j,m}-\theta_{j,m}\|$ and $\frac{1}{N^2}\sum_{i,j=1}^{N}|\hat{\mu}^i_{j,m}-\mu_{j,m}|$ asymptotically converge to zero, which aligns with the theoretical result established in Theorem 2.

D. Ablation experiment

Note that in practice, we adopt the diminishing learning rate $\eta_{\theta,m} \sim \mathcal{O}(\frac{\eta_\theta}{m})$ as required by Assumption 7, where η_θ denotes the initial learning rate. Fig. 4 presents the objective return $F(\boldsymbol{\theta})$ and average constraint return $\bar{G}(\boldsymbol{\theta})$ under different initial learning rates $\eta_\theta \in \{0.005, 0.02, 0.05, 0.15, 0.30\}$. Both too small an initial rate (e.g., $\eta_\theta = 0.005$) that converges slowly to a lower plateau and too large an initial rate (e.g., $\eta_\theta = 0.30$) that leads to instability result in degraded objective performance. An intermediate rate $\eta_\theta \in \{0.02, 0.05, 0.15\}$ achieves a good balance between convergence speed, objective performance, and constraint satisfaction.



(a) Objective return (b) Average constraint return

Fig. 4. The evolution of the objective return and average constraint return of DSPD algorithm under different learning rates.

VII. Conclusions

In this paper, we study the CMARL problem under coupled policies in a distributed setting without global environmental information. We propose a DSPD algorithm that operates solely based on a prescribed local neighborhood information, and establish theoretical guarantees showing that the proposed algorithm achieves an ϵ-policy stationary convergence with an approximation error of $\mathcal{O}(\gamma^{\frac{\kappa+1}{\kappa_p}})$. Simulation results on a wireless access-control network validate the effectiveness of the proposed algorithm.

In future research, several important directions remain open: (i) improving the convergence rate by designing more efficient learning rate schedules that simultaneously handle the coupled policy structure and distributed estimation errors, with the goal of achieving a convergence rate comparable to that of existing independent policy-based algorithms; (ii) incorporating formal privacy mechanisms such as differential privacy into the time-varying learning network to enforce rigorous privacy guarantees on transmitted parameters; (iii) extending the framework to handle truly lossy communication by replacing the uniformly strongly connected assumption with a probabilistic connectivity condition, and (iv) extending the proposed framework to more complex settings such as heterogeneous autonomous swarm environments with large-scale agents.

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VIII. Analysis of Main Results

A. Analysis of Theorem 1

To facilitate a clear presentation of the proof for Theorem 1, we introduce some definitions of auxiliary functions and the corresponding lemmas.

Recalling the definitions of $Q_i^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})$ in (11) and $h(\kappa,\kappa_p)$ in (30), we establish the exponential decay property of CMARL under coupled policies as follows.

Lemma 2: The CMARL problem under coupled policies satisfies the (ϑ,ϱ)-exponential decay property, i.e., for any joint policy $\boldsymbol{\pi_\theta}$, Lagrangian multiplier $\boldsymbol{\mu}$, agent $i\in\mathcal{N}$, $\kappa\geq 0$, $s_{\mathcal{N}_i^{E,\kappa}}\in\mathcal{S}_{\mathcal{N}_i^{E,\kappa}}$, $a_{\mathcal{N}_i^{E,\kappa}}\in\mathcal{A}_{\mathcal{N}_i^{E,\kappa}}$, $s_{\mathcal{N}_{-i}^{E,\kappa}},s'_{\mathcal{N}_{-i}^{E,\kappa}}\in\mathcal{S}_{\mathcal{N}_{-i}^{E,\kappa}}$, and $a_{\mathcal{N}_{-i}^{E,\kappa}},a'_{\mathcal{N}_{-i}^{E,\kappa}}\in\mathcal{A}_{\mathcal{N}_{-i}^{E,\kappa}}$, the local Lagrangian Q-function $Q_i^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})$ satisfies

$$\Big|Q_i^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}};\boldsymbol{\mu}) - Q_i^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},s'_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}},a'_{\mathcal{N}_{-i}^{E,\kappa}};\boldsymbol{\mu})\Big|\leq\vartheta\varrho^{\kappa+1},\quad(33)$$

where $\vartheta=\frac{2(R_f+\mu_{max}R_g)}{(1-\gamma)\gamma^{\kappa-h(\kappa,\kappa_p)}}$ and $\varrho=\gamma$.

The detailed proof of Lemma 2 is provided in Section IX-B. Lemma 2 shows that each agent i can approximate its local Lagrangian Q-function $Q_i^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})$ using only the state-action pairs $(s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}})$ from its κ-hop neighbors, and the approximation error decays exponentially as κ increases. While this result shares a similar spirit with Lemma 3 in [19], the coupled policy structure introduces a key difference: under independent policies in [19], the decay exponent is $\kappa+1$, whereas under coupled policies, the policy coupling accelerates the propagation of out-of-neighborhood differences through the κ_p-hop neighborhood, yielding a tighter decay exponent $h(\kappa,\kappa_p)+1$.

Recalling the definition of $d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s})$ in (14), we define $\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a})$ as the discounted state-action visitation distribution of $(\boldsymbol{s},\boldsymbol{a})\in\boldsymbol{\mathcal{S}}\times\boldsymbol{\mathcal{A}}$, which is given by

$$\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a})=d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s})\boldsymbol{\pi_\theta}(\boldsymbol{a}|\boldsymbol{s}).\quad(34)$$

Inspired by the exponential decay property in Lemma 2 and the definition of $\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a})$ in (34), we design a class of truncated Lagrangian Q-functions as

$$\begin{aligned}&Q_{tru,i}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}};\boldsymbol{\mu})\\=&\sum_{s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}}}\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}}|s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}})\\&\times Q_i^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}};\boldsymbol{\mu}),\end{aligned}\quad(35)$$

where $\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}}|s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}})$ denotes the weight coefficient and is represented as

$$\begin{aligned}&\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}}|s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}})\\=&\frac{\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}})}{\sum_{s'_{\mathcal{N}_{-i}^{E,\kappa}},a'_{\mathcal{N}_{-i}^{E,\kappa}}}\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},s'_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}},a'_{\mathcal{N}_{-i}^{E,\kappa}})}.\end{aligned}\quad(36)$$

By definition of $\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}}|s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}})$ in (36) and Assumption 4, it is obvious that the weight coefficient $\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}}|s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}})$ is non-negative and satisfies

$$\sum_{s'_{\mathcal{N}_{-i}^{E,\kappa}},a'_{\mathcal{N}_{-i}^{E,\kappa}}}\xi_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(s'_{\mathcal{N}_{-i}^{E,\kappa}},a'_{\mathcal{N}_{-i}^{E,\kappa}}|s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}})=1.\quad(37)$$

Based on Lemma 2 and (37), we further obtain

$$\begin{aligned}&\Big|Q_{tru,i}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}};\boldsymbol{\mu})-Q_i^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})\Big|\\\leq&\frac{2(R_f+\mu_{max}R_g)}{1-\gamma}\gamma^{h(\kappa,\kappa_p)+1}.\end{aligned}\quad(38)$$

Based on the definition of $Q_{tru,i}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}};\boldsymbol{\mu})$ in (35), we can derive the following lemma.

Lemma 3: For any agent $i\in\mathcal{N}$, $l\in\mathcal{N}_{-i}^{E,\kappa+2\kappa_p}$, joint policy $\boldsymbol{\pi_\theta}$, and Lagrangian multiplier $\boldsymbol{\mu}$, we have

$$\begin{aligned}&\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[Q_{tru,l}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_l^{E,\kappa}},a_{\mathcal{N}_l^{E,\kappa}};\boldsymbol{\mu})\\&\times\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big]\\=&0.\end{aligned}\quad(39)$$

The detailed proof of Lemma 3 can be found in Section IX-C. Based on Lemma 3, the proof of Theorem 1 is presented as follows.

Proof. By the definitions of $\widehat{Q_i^{\boldsymbol{\pi_\theta}}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})$ in (17) and $g_{app,i}^{\boldsymbol{\pi_\theta},\boldsymbol{\mu}}$ in (18), we directly have

$$\begin{aligned}g_{app,i}^{\boldsymbol{\pi_\theta},\boldsymbol{\mu}}=&\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[\frac{1}{N}\sum_{l\in\mathcal{N}_i^{\kappa+2\kappa_p}}Q_l^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})\\&\times\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big].\end{aligned}\quad(40)$$

By using (40) and the definition of $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ in (16), we further have

$$\Big\|g_{app,i}^{\boldsymbol{\pi_\theta},\boldsymbol{\mu}}-\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})\Big\|$$

$$=\Big\|\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[\frac{1}{N}\sum_{l\in\mathcal{N}_{-i}^{E,\kappa+2\kappa_p}}Q_l^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}) \times\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big]\Big\| \tag{41}$$

$$\leq\Big\|\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[\frac{1}{N}\sum_{l\in\mathcal{N}_{-i}^{E,\kappa+2\kappa_p}}\Big(Q_l^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu}) - Q_{tru,l}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_l^{E,\kappa}},a_{\mathcal{N}_l^{E,\kappa}};\boldsymbol{\mu})\Big) \times\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big]\Big\| + \Big\|\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[\frac{1}{N}\sum_{l\in\mathcal{N}_{-i}^{E,\kappa+2\kappa_p}}Q_{tru,l}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_l^{E,\kappa}},a_{\mathcal{N}_l^{E,\kappa}};\boldsymbol{\mu}) \times\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big]\Big\| \tag{42}$$

$$\leq\frac{2(R_f+\mu_{max}R_g)BN}{(1-\gamma)^2}\gamma^{h(\kappa,\kappa_p)+1}, \tag{43}$$

where (42) follows from the triangle inequality of $\mathcal{L}_2$-norm, and the last inequality comes from Assumption 5, (38), and (39). □

B. Analysis of Theorem 2

Before presenting the proof of Theorem 2, we first introduce an important lemma that will be used throughout the proof.

Lemma 4: [17] Suppose Assumption 6 holds. For each integer $k\geq 1$, there is a stochastic vector sequence $\{\phi_t\}_{t\geq 1}$ such that for all $i,j\in\mathcal{N}$ and $t\geq k$,

$$|W_{ij,t:k}^L-\phi_{i,t}|\leq M_1\lambda^{t-k}, \tag{44}$$

where $M_1>0$, $\lambda\in(0,1)$, $W_{t:k}^L=W_t^LW_{t-1}^L\cdots W_k^L$, and $W_{ij,t:k}^L$ being the i-th row and the j-th column element in $W_{t:k}^L$, and $\phi_{i,t}$ represents the i-th element in ϕ_t.

Lemma 4 indicates that there exists $w_{max}>1$, such that $\|W_{t:k}^L\|\leq w_{max}$ for all $t\geq k$.

In what follows, we provide the detailed proof of Theorem 2.

Proof. For Case (i), by the definition of $\hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k)$ in (20), we have that

$$\|\hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k)\|=\Big\|\frac{1}{N}\Big(\sum_{t=0}^{T_1(k)}\gamma^{t/2}g_{i,t}-c_i\Big)\Big\| \leq\frac{R_g+c_{max}}{(1-\gamma^{1/2})N}=L_1^{\mu}, \tag{45}$$

where the inequality comes from Assumption 3 and holds for all $m\geq 1$.

Next, we will demonstrate the convergence of $\hat{\mu}_{j,m}^i$. For the convenience of expression, we define $\breve{\mu}_{j,m}=\big(\breve{\mu}_{j,m}^1,\cdots,\breve{\mu}_{j,m}^N\big)^\top\in\mathbb{R}^N$, based on (23), the update of $\breve{\mu}_{j,m}$ can be written as

$$\breve{\mu}_{j,m+1}=W_m^L\Big(\breve{\mu}_{j,m}+N\big(\mu_{j,m+1}-\mu_{j,m}\big)e_j\Big), \tag{46}$$

where e_j is the unit vector with the j-th element is 1 and other elements are 0. Since W_m^L is the column stochastic matrix, we multiply both sides of (46) by $(1/N)\mathbf{1}_N^\top$ and have

$$\frac{1}{N}\mathbf{1}_N^\top\breve{\mu}_{j,m+1}=\frac{1}{N}\mathbf{1}_N^\top\Big(\breve{\mu}_{j,m}+N\big(\mu_{j,m+1}-\mu_{j,m}\big)e_j\Big) =\frac{1}{N}\mathbf{1}_N^\top\breve{\mu}_{j,m}+\big(\mu_{j,m+1}-\mu_{j,m}\big). \tag{47}$$

From (47), we further have

$$\frac{1}{N}\mathbf{1}_N^\top\breve{\mu}_{j,m+1}-\mu_{j,m+1}=\frac{1}{N}\mathbf{1}_N^\top\breve{\mu}_{j,m}-\mu_{j,m}=0, \tag{48}$$

where the last equality can be obtained from the initial setting of $\breve{\mu}_{j,1}^i=\mu_{j,1}=0$ on Line 2 of Algorithm 1.

Denote $\delta_{m+1}=W_m^LN\big(\mu_{j,m+1}-\mu_{j,m}\big)e_j$, (46) can be further written as

$$\breve{\mu}_{j,m+1}=W_m^L\breve{\mu}_{j,m}+\delta_{m+1} =W_{m:1}^L\breve{\mu}_{j,1}+\sum_{k=2}^{m}W_{m:k}^L\delta_k+\delta_{m+1}, \tag{49}$$

where $W_{m:k}^L=W_m^LW_{m-1}^L\cdots W_k^L,\forall m\geq k$. Multiplying W_{m+1}^L on both sides of (49), we have

$$W_{m+1}^L\breve{\mu}_{j,m+1}=W_{m+1:1}^L\breve{\mu}_{j,1}+\sum_{k=2}^{m+1}W_{m+1:k}^L\delta_k. \tag{50}$$

Given that both W_m^L and $W_{m:k}^L$ are column stochastic matrices, we multiply both sides of (50) by $\mathbf{1}_N^\top$ and have

$$\mathbf{1}_N^\top\breve{\mu}_{j,m+1}=\mathbf{1}_N^\top\breve{\mu}_{j,1}+\sum_{k=2}^{m+1}\mathbf{1}_N^\top\delta_k. \tag{51}$$

Subtracting (51) multiplied by the vector $\phi_m\in\mathbb{R}^N$ in Lemma 4 from (50), we obtain

$$(W_{m+1}^L-\phi_{m+1}\mathbf{1}_N^\top)\breve{\mu}_{j,m+1} =(W_{m+1:1}^L-\phi_{m+1}\mathbf{1}_N^\top)\breve{\mu}_{j,1}+\sum_{k=2}^{m+1}(W_{m+1:k}^L-\phi_{m+1}\mathbf{1}_N^\top)\delta_k. \tag{52}$$

Denote $\xi_{m+1:k}^L=W_{m+1:k}^L-\phi_{m+1}\mathbf{1}_N^\top\in\mathbb{R}^{N\times N}$, (52) can be written as

$$W_{m+1}^L\breve{\mu}_{j,m+1}=\phi_{m+1}\mathbf{1}_N^\top\breve{\mu}_{j,m+1}+\xi_{m+1:1}^L\breve{\mu}_{j,1} +\sum_{k=2}^{m+1}\xi_{m+1:k}^L\delta_k. \tag{53}$$

Recalling the update (19a) with $p_{i,1}=1$ for all $i\in\mathcal{N}$, we let $P_m=(p_{1,m},\cdots,p_{N,m})^\top\in\mathbb{R}^N$ and directly have that $P_{m+1}=W_m^LP_m=W_{m:1}^LP_1$ and $\mathbf{1}_N^\top P_{m+1}=\mathbf{1}_N^\top P_m=\mathbf{1}_N^\top P_1=N$. Furthermore, we have

$$P_{m+1}-\phi_m\mathbf{1}_N^\top P_{m+1}=(W_{m:1}^L-\phi_m\mathbf{1}_N^\top)P_1$$

and

$$P_{m+1}=N\phi_m+\xi_{m:1}^L\mathbf{1}_N. \tag{54}$$

Let $\xi_{i,m:k}^L\in\mathbb{R}^N$ be the i-th row element of $\xi_{m:k}^L$. By using (53) and (54), (19b) can derive

$$\Big|\hat{\mu}_{j,m}^i-\frac{1}{N}\mathbf{1}_N^\top\breve{\mu}_{j,m}\Big| =\left|\frac{\phi_{i,m}\mathbf{1}_N^\top\breve{\mu}_{j,m}+\xi_{i,m:1}^L\breve{\mu}_{j,1}+\sum_{k=2}^m\xi_{i,m:k}^L\delta_k}{N\phi_{i,m}+\xi_{i,m:1}^L\mathbf{1}_N}-\frac{\mathbf{1}_N^\top\breve{\mu}_{j,m}}{N}\right| =\left|\frac{N\xi_{i,m:1}^L\breve{\mu}_{j,1}+N\sum_{k=2}^m\xi_{i,m:k}^L\delta_k-\xi_{i,m:1}^L\mathbf{1}_N\mathbf{1}_N^\top\breve{\mu}_{j,m}}{N(N\phi_{i,m}+\xi_{i,m:1}^L\mathbf{1}_N)}\right|$$

$$\leq \frac{\|\xi^L_{i,m:1}\|\|\breve{\mu}_{j,1}\| + \|\sum_{k=2}^m \xi^L_{i,m:k}\delta_k\|}{N\phi_{i,m} + \xi^L_{i,m:1}\mathbf{1}_N} + \frac{\|\xi^L_{i,m:1}\mathbf{1}_N\|\|\mathbf{1}_N^\top \breve{\mu}_{j,m}\|}{N(N\phi_{i,m} + \xi^L_{i,m:1}\mathbf{1}_N)} \tag{55}$$

$$\leq \frac{\sqrt{N}\big(M_1\lambda^{m-1}\|\breve{\mu}_{j,1}\| + \sum_{k=2}^m M_1\lambda^{m-k}\|\delta_k\|\big)}{\zeta} + \frac{M_1\lambda^{m-1}\|\breve{\mu}_{j,m}\|}{\zeta}, \tag{56}$$

where the first inequality uses the fact that $|a^\top b| \leq \|a\|\|b\|, \forall a, b \in \mathbb{R}^N$ and the last inequality comes from Lemma 4 and the observation that $N\phi_{i,m} + \xi^L_{i,m:1}\mathbf{1}_N \geq \zeta$, a result established in the proof of Lemma 1 in [17].

For $\lambda^{m-1}\|\breve{\mu}_{j,m}\|$ in the right side of (56), we can get

$$\lambda^{m-1}\|\breve{\mu}_{j,m}\| = \lambda^{m-1}\Big\|W^L_{m-1:1}\breve{\mu}_{j,1} + \sum_{k=2}^{m-1} W^L_{m-1:k}\delta_k + \delta_m\Big\|$$

$$\leq \lambda^{m-1}\sum_{k=2}^{m-1} w_{max}\|\delta_k\| + \|\delta_m\| \tag{57}$$

$$\leq w_{max}\sum_{k=2}^{m}\lambda^{m-k}\|\delta_k\|, \tag{58}$$

where the first inequality (57) can be obtained by the initial setting of $\breve{\mu}^i_{j,1} = 0, \forall i, j \in \mathcal{N}$ on Line 2 of Algorithm 1 and $\|W^L_{t:k}\| \leq w_{max}$ for all $t \geq k$ claimed after Lemma 4, and the last inequality comes from that $\lambda \in (0, 1)$.

For $\|\delta_k\|$ with $2 \leq k \leq m$ in the right side of (56), by using $\delta_k = W^L_{k-1}N\big(\mu_{j,k} - \mu_{j,k-1}\big)e_j$, we have

$$\|\delta_k\| = \|W^L_{k-1}N\big(\mu_{j,k} - \mu_{j,k-1}\big)e_j\|$$

$$\leq N\|W^L_{k-1}\|\|\mu_{j,k} - \mu_{j,k-1}\|$$

$$= N\|W^L_{k-1}\|\|P_{\mathcal{U}_j}(\mu_{j,k-1} - \eta_{\mu,k-1}\bar{h}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_{k-1}},\hat{\boldsymbol{\mu}}_{k-1}}_{app,j}) - \mu_{j,k-1}\| \tag{59}$$

$$\leq w_{max}N\eta_{\mu,k-1}\|\bar{h}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_{k-1}},\hat{\boldsymbol{\mu}}_{k-1}}_{app,j}\| \tag{60}$$

$$\leq w_{max}L^\mu_1 N\eta_{\mu,k-1}, \tag{61}$$

where (59) uses the update (22), (60) can be obtained by the projection inequality, and the last inequality (61) can be obtained by (45) and (21).

Substituting (58), (61), and the fact that $\breve{\mu}^i_{j,1} = 0, \forall i, j \in \mathcal{N}$ into (56), we can have

$$\Big|\hat{\mu}^i_{j,m} - \mu_{j,m}\Big| \leq \frac{(\sqrt{N} + w_{max})M_1}{\zeta}\sum_{k=2}^m \lambda^{m-k}\|\delta_k\| \leq \frac{(\sqrt{N} + w_{max})w_{max}L^\mu_1 M_1 N}{\zeta}\sum_{k=2}^m \lambda^{m-k}\eta_{\mu,k-1}. \tag{62}$$

Taking limits on m on both sides of (62), we further have

$$\lim_{m\to\infty}|\hat{\mu}^i_{j,m} - \mu_{j,m}| \leq \lim_{m\to\infty}\frac{(\sqrt{N} + w_{max})w_{max}L^\mu_1 M_1 N}{\zeta}\sum_{k=2}^m \lambda^{m-k}\eta_{\mu,k-1}, \tag{63}$$

$$= 0 \tag{64}$$

with rate $\mathcal{O}(\frac{1}{m})$, where the last inequality can be achieved by $\eta_{\mu,m} \sim \mathcal{O}(\frac{1}{m})$ in Assumption 7 and Lemma 3.1 in [30]. Hence, we can prove the Case (i).

For Case (ii), by the definition of $\hat{g}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}_{app,i}(k)$ in (24), we have

$$\|\hat{g}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}_{app,i}(k)\| = \frac{1}{1-\gamma}\Big\|\hat{Q}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}_{i,T_2(k)}\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_{j,T_2(k)}|s_{\mathcal{N}_j^{E,\kappa_p},T_2(k)}, \hat{\theta}^i_{j,m}, \hat{\theta}^i_{\mathcal{N}^{E,\kappa_p}_{j,-j},m})\Big\|$$

$$\leq \frac{1}{1-\gamma}\Big|\frac{1}{N}\sum_{t=0}^{T_3(k)}\gamma^{t/2}\sum_{j\in\mathcal{N}_i^{E,\kappa+2\kappa_p}}(f_{j,T_2(k)+t} + \hat{\mu}^i_{j,m+1}g_{j,T_2(k)+t})\Big| \times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\Big\|\nabla_{\theta_i}\log\pi_j(a_{j,T_2(k)}|s_{\mathcal{N}_j^{E,\kappa_p},T_2(k)}, \hat{\theta}^i_{j,m}, \hat{\theta}^i_{\mathcal{N}^{E,\kappa_p}_{j,-j},m})\Big\|, \tag{65}$$

where the last inequality (65) can be obtained by the definition of $\hat{Q}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}_{i,T_2(k)}$ in (25) and triangular inequality.

Since (45) holds for any joint policy, the conclusion $\lim_{m\to\infty}\hat{\mu}^i_{j,m} = \mu_{j,m}$ can be consistently upheld throughout the learning process. Based on this fact and $\mu_{j,m} \in \mathcal{U}_j = [0, \mu_{max}]$, we have that $\hat{\mu}^i_{j,m}$ is also bounded, i.e., there exists $\tilde{\mu}_{max} \geq \mu_{max} > 0$ such that

$$|\hat{\mu}^i_{j,m}| \leq \tilde{\mu}_{max}, \forall i, j \in \mathcal{N}, m \geq 1. \tag{66}$$

Substituting (66) into (65), we have

$$\|\hat{g}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}_{app,i}(k)\| \leq \frac{BN(R_f + \tilde{\mu}_{max}R_g)}{(1-\gamma)(1-\gamma^{1/2})} = L^\theta_1, \tag{67}$$

where the inequality (67) can be achieved by Assumptions 3-5. Similar to the proof of Case (i), by employing (67) and $\eta_{\theta,m} \sim \mathcal{O}(\frac{1}{\sqrt{m}})$ in Assumption 7, we can obtain $\lim_{m\to\infty}\hat{\theta}^i_{j,m} = \theta_{j,m}$ with rate $\mathcal{O}(\frac{1}{\sqrt{m}})$ for all $i, j \in \mathcal{N}$. Given that the proof procedure is strikingly analogous to that of Case (i), it is omitted for brevity. □

C. Analysis of Theorem 3

To clearly present the progress of the proof for Theorem 3, we introduce several key definitions, along with intermediate lemmas and corollaries.

Recalling $Q^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}(\boldsymbol{s}, \boldsymbol{a}; \boldsymbol{\mu})$ in (10), $Q^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}_{g,i}(\boldsymbol{s}, \boldsymbol{a})$ in (13), and $\widehat{Q^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}_i}(\boldsymbol{s}, \boldsymbol{a}; \boldsymbol{\mu})$ in (17), we define $Q^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}(\boldsymbol{s}, \boldsymbol{a}; \hat{\boldsymbol{\mu}}^i_{m+1})$ and $\widehat{Q^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}_i}(\boldsymbol{s}, \boldsymbol{a}; \hat{\boldsymbol{\mu}}^i_{m+1})$ as the global Lagrangian Q-function and the neighbors' averaged Lagrangian Q-function, respectively, associated with the joint policy $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$ and the Lagrange multiplier $\hat{\boldsymbol{\mu}}^i_{m+1}$. Moreover, we denote by $Q^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}_{g,i}(\boldsymbol{s}, \boldsymbol{a})$ the local Q-function corresponding to the constraint reward under the same joint policy.

On the basis of these definitions, we define $\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\mu}}_m)$ as the Lagrangian gradient and $\nabla_{\theta_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\mu}}_{m+1})$ as the policy gradient of agent i under the executed joint policy $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$, Lagrangian multipliers $\hat{\boldsymbol{\mu}}_m$ and $\hat{\boldsymbol{\mu}}_{m+1}$, respectively. They are described as follows:

$$\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\mu}}_m) = \frac{1}{N}\Big(\mathbb{E}_{\boldsymbol{s}\sim\boldsymbol{\rho},\boldsymbol{a}\sim\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}\Big[Q^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}_{g,i}(\boldsymbol{s}, \boldsymbol{a})\Big] - c_i\Big) \tag{68}$$

and

$$\begin{aligned}&\nabla_{\theta_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_{m+1})\\=&\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}},\boldsymbol{a}\sim\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}\Big[Q^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}(\boldsymbol{s},\boldsymbol{a};\hat{\boldsymbol{\mu}}_{m+1}^i)\\&\times\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa}},\hat{\theta}_{j,m}^i,\hat{\theta}^i_{\mathcal{N}_{j,-j}^{E,\kappa_p},m})\Big].\end{aligned}\tag{69}$$

Similar to the approximated policy gradient $g_{app,i}^{\boldsymbol{\pi}_{\boldsymbol{\theta}},\boldsymbol{\mu}}$ in (18), we define $g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ as the approximated coupled policy gradient of agent i under the executed joint policy $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$ and Lagrangian multiplier $\hat{\boldsymbol{\mu}}_{m+1}$, which is described as

$$\begin{aligned}g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}=&\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}},\boldsymbol{a}\sim\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}\Big[\widehat{Q_i^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}}(\boldsymbol{s},\boldsymbol{a};\hat{\boldsymbol{\mu}}_{m+1}^i)\\&\times\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\hat{\theta}_{j,m}^i,\hat{\theta}^i_{\mathcal{N}_{j,-j}^{E,\kappa_p},m})\Big].\end{aligned}\tag{70}$$

Having introduced the auxiliary definitions (68)-(70), we now present several key lemmas and corollaries that form the theoretical foundation of the analysis of Theorem 3.

Lemma 5 (Unbiased gradient estimates for (20) and (24)): Suppose Assumptions 1-7 hold. In each m-th iteration, the Lagrangian gradient estimation $\hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k)$ and the approximated policy gradient estimation $\hat{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}(k)$ satisfy

(i) $\hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k)$ is an unbiased estimate of $\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_m)$, i.e., $\mathbb{E}_{T_1(k)}\Big[\hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k)\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]=\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_m)$ for all $k\in\{1,\cdots,K_\mu\}$;

(ii) $\hat{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}(k)$ is an unbiased estimate of $g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$, i.e., $\mathbb{E}_{T_2(k),T_3(k)}\Big[\hat{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}(k)\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]=g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ for all $k\in\{1,\cdots,K_\theta\}$.

The proof of Lemma 5 is provided in Section IX-D. Lemma 5 establishes that each agent i can achieve unbiased estimates of the Lagrangian gradient $\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_m)$ and the approximated policy gradient $g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ through samples $\{\boldsymbol{s},\boldsymbol{a}\}_{0:T_1(k)}$ and $\{\boldsymbol{s},\boldsymbol{a}\}_{0:T_2(k)+T_3(k)}$, respectively. From Lemma 5, the following corollary immediately follows.

Corollary 1: Suppose Assumptions 1-7 hold. In each m-th iteration, for any $\delta\in(0,1)$, the following inequalities hold with probability at least $1-\delta$:

$$\big\|\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}-\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_m)\big\|\leq 2L_1^\mu\sqrt{\frac{\log(2/\delta)}{2K_\mu}}\tag{71}$$

and

$$\big\|\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}-g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\big\|\leq 2L_1^\theta\sqrt{\frac{\log(2/\delta)}{2K_\theta}}.\tag{72}$$

Proof. By Theorem 2, Lemma 5, and Hoeffding inequality, (71) and (72) can be obtained directly. □

Corollary 1 establishes the upper bound of error between $\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}$ in (22) and $\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_m)$ in (68), along with the error between $\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ in (27) and $g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ in (70).

In order to further quantify the discrepancy between $\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}$ and the true Lagrangian gradient $\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_m)$ as well as the discrepancy between $\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ and true policy gradient $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})$, we define $L_2^\mu=BR_g/(1-\gamma)^2$, $L_2^{\theta\mu}=B\sqrt{N}R_g/(1-\gamma)^2$, $L_2^{\theta\theta}=LN(R_f+\tilde{\mu}_{max}R_g)/(1-\gamma)^2+(1+\gamma)B^2N^2(R_f+\tilde{\mu}_{max}R_g)/(1-\gamma)^3$, and derive the following lemma.

Lemma 6: Suppose Assumptions 3-5 hold. For any agent $i\in\mathcal{N}$ and $m\geq 1$, we have

(i) $\big\|\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_m)-\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_m)\big\|\leq L_2^\mu\big\|\mathbf{1}_N\otimes\boldsymbol{\theta}_m-\hat{\boldsymbol{\theta}}_m\big\|$;

(ii) $\big\|\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})-\nabla_{\theta_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_{m+1})\big\|\leq L_2^{\theta\mu}\big\|\boldsymbol{\mu}_{m+1}-\hat{\boldsymbol{\mu}}_{m+1}^i\big\|+L_2^{\theta\theta}\big\|\mathbf{1}_N\otimes\boldsymbol{\theta}_m-\hat{\boldsymbol{\theta}}_m\big\|$.

The detailed proof of Lemma 6 is provided in Section IX-E. Lemma 6 indicates that the upper bounds on the error between $\nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_m)$ and the true Lagrangian gradient $\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_m)$ depends on the discrepancy between the estimated policy parameters $\{\hat{\boldsymbol{\theta}}_m^i\}_{i\in\mathcal{N}}$ and the true policy parameter $\boldsymbol{\theta}_m$. In contrast, the upper bounds on the error between $\nabla_{\theta_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_{m+1})$ and the true policy gradient $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})$ depend not only on the aforementioned discrepancy but also on the discrepancy between the estimated Lagrangian multipliers $\{\hat{\boldsymbol{\mu}}_{m+1}^i\}_{i\in\mathcal{N}}$ and the true Lagrangian multiplier $\boldsymbol{\mu}_{m+1}$.

Define $\epsilon_\mu(m,K_\mu)=L_2^\mu\sqrt{N}\big\|\mathbf{1}_N\otimes\boldsymbol{\theta}_m-\hat{\boldsymbol{\theta}}_m\big\|+2L_1^\mu\sqrt{N\log(2/\delta)/2K_\mu}$, $\epsilon_\theta(m,K_\theta)=L_2^{\theta\mu}\sqrt{N}\big\|\mathbf{1}_N\otimes\boldsymbol{\mu}_{m+1}-\hat{\boldsymbol{\mu}}_{m+1}\big\|+L_2^{\theta\theta}\sqrt{N}\big\|\mathbf{1}_N\otimes\boldsymbol{\theta}_m-\hat{\boldsymbol{\theta}}_m\big\|+2L_1^\theta\sqrt{N\log(2/\delta)/2K_\theta}$, and $\epsilon_0(\kappa,\kappa_p)=2(R_f+\tilde{\mu}_{max}R_g)BN^{\frac{3}{2}}\gamma^{h(\kappa,\kappa_p)+1}/(1-\gamma)^2$. By combining Theorem 1, Corollary 1, and Lemma 6, we have the following corollary.

Corollary 2: Suppose Assumptions 3-5 hold. For any $i\in\mathcal{N}$ and $t\geq 1$, we have

(i) $\big\|\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_m)-\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}\big\|\leq\epsilon_\mu(m,K_\mu)/\sqrt{N}$;

(ii) $\big\|\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})-\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\big\|\leq\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)/\sqrt{N}$.

The detailed proof of Corollary 2 can be found in Section IX-F. Corollary 2 establishes the the upper bounds of the error between $\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_m)$ and $\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}$, as well as the upper bounds of the error between $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})$ and $\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$. It is important to note that in Case (i), the upper bound of the error is determined by the discrepancy between $\{\hat{\boldsymbol{\theta}}_m^i\}_{i\in\mathcal{N}}$ and $\boldsymbol{\theta}_m$, as well as the number of sample batches K_μ. Conversely, in Case (ii), the upper bound of the error is influenced by the errors between $\{\hat{\boldsymbol{\theta}}_m^i\}_{i\in\mathcal{N}}$ and $\boldsymbol{\theta}_m$, the errors between $\{\hat{\boldsymbol{\mu}}_{m+1}^i\}_{i\in\mathcal{N}}$ and $\boldsymbol{\mu}_{m+1}$, the number of sample batches K_μ, and the truncated distance κ.

Recall the definition of $L_{\theta\theta}$ in Theorem 3. We denote $\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})=\big(\nabla_{\theta_1}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})^\top,\cdots,\nabla_{\theta_N}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})^\top\big)^\top$ and can derive the following corollary.

Corollary 3: Suppose Assumptions 3-5 hold, we have that $\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ is Lipschitz continuous with respect to $\boldsymbol{\theta}$, i.e., $\|\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})-\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}',\boldsymbol{\mu})\|\leq L_{\theta\theta}\|\boldsymbol{\theta}-\boldsymbol{\theta}'\|$ for any $\boldsymbol{\theta},\boldsymbol{\theta}'\in\boldsymbol{\Theta}$.

Proof. The proof of this corollary follows a similar process to that of Lemma 6. Only note that the primary difference lies in the variation of coefficients. For the sake of brevity, the detailed steps are omitted here. □

Corollary 3 shows the $L_{\theta\theta}$-smoothness of $\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ with respect to $\boldsymbol{\theta}$. It is a direct extension of Lemma 6 and serves as the foundation for analyzing the first-order stationary convergence of Algorithm 1.

Recall Theorem 2 states that for all $i,j\in\mathcal{N}$, as $m\to\infty$, $\hat{\mu}_{j,m}^i$ and $\hat{\theta}_{j,m}^i$ converge to $\mu_{j,m}$ and $\theta_{j,m}$, respectively. By

leveraging this observation and Corollaries 2-3, the proof of Theorem 3 is as follows.

Proof. Recalling the definitions of $\bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}$ in (21) and $\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}$ in (26), we denote $\bar{\boldsymbol{h}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m} = \big((\bar{h}_{app,1}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m})^\top,\cdots,(\bar{h}_{app,N}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m})^\top\big)^\top$ and $\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}} = \big((\bar{g}_{app,1}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}})^\top,\cdots,(\bar{g}_{app,N}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}})^\top\big)^\top$.

Consider that the feasible set Θ_i is convex for all $i\in\mathcal{N}$, by the update of the policy parameter in (27) and the property of the projection operator, it holds that

$$\langle \boldsymbol{\theta}_m+\eta_{\theta,m}\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}-\boldsymbol{\theta}_{m+1},\boldsymbol{\theta}-\boldsymbol{\theta}_{m+1}\rangle\le 0,\forall\boldsymbol{\theta}\in\boldsymbol{\Theta}.$$

Then, we have

$$\langle \bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}-\boldsymbol{\theta}_{m+1}\rangle\le\frac{1}{\eta_{\theta,m}}\langle\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m,\boldsymbol{\theta}-\boldsymbol{\theta}_{m+1}\rangle \tag{73}$$

for all $\boldsymbol{\theta}\in\boldsymbol{\Theta}$. According to the definition of $\mathcal{Y}(\boldsymbol{\theta},\boldsymbol{\mu})$ in (9), we have

$$\begin{aligned}
&[\mathcal{Y}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})]^2\\
=&\Big[\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\|\boldsymbol{\theta}-\boldsymbol{\theta}_m\|\le1}\langle\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1}),\boldsymbol{\theta}-\boldsymbol{\theta}_m\rangle\Big]^2\\
\le&2\Big[\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\|\boldsymbol{\theta}-\boldsymbol{\theta}_m\|\le1}\langle\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}-\boldsymbol{\theta}_m\rangle\Big]^2\\
&+2\big\|\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})-\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\big\|^2\\
\le&2\Big[\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\|\boldsymbol{\theta}-\boldsymbol{\theta}_m\|\le1}\langle\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}-\boldsymbol{\theta}_m\rangle\Big]^2\\
&+2\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2,
\end{aligned}\tag{74}$$

where the last inequality can be obtained by Case (ii) in Corollary 2. For $\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\|\boldsymbol{\theta}-\boldsymbol{\theta}_m\|\le1}\langle\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}-\boldsymbol{\theta}_m\rangle$ on the right side of (74), we have

$$\begin{aligned}
&\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\|\boldsymbol{\theta}-\boldsymbol{\theta}_m\|\le1}\langle\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}-\boldsymbol{\theta}_m\rangle\\
=&\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\|\boldsymbol{\theta}-\boldsymbol{\theta}_m\|\le1}\big\{\langle\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}-\boldsymbol{\theta}_{m+1}\rangle\\
&+\langle\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\rangle\big\}\\
\le&\max_{\boldsymbol{\theta}\in\boldsymbol{\Theta},\|\boldsymbol{\theta}-\boldsymbol{\theta}_m\|\le1}\Big\{\frac{1}{\eta_{\theta,m}}\langle\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m,\boldsymbol{\theta}-\boldsymbol{\theta}_m\rangle\\
&+\frac{1}{\eta_{\theta,m}}\langle\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m,\boldsymbol{\theta}_m-\boldsymbol{\theta}_{m+1}\rangle\\
&+\big\|\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\big\|\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|\Big\}
\end{aligned}\tag{75}$$

$$\begin{aligned}
\le&\frac{1}{\eta_{\theta,m}}\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\|-\frac{1}{\eta_{\theta,m}}\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\|^2\\
&+\big\|\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\big\|\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|\\
\le&\Big(\frac{1}{\eta_{\theta,m}}+\sqrt{N}L_1^\theta\Big)\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|,
\end{aligned}\tag{76}$$

where (75) can be obtained by (73) and (76) uses (67). Substituting (76) into (74), we have

$$\begin{aligned}
[\mathcal{Y}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})]^2\le&2\Big(\frac{1}{\eta_{\theta,m}}+\sqrt{N}L_1^\theta\Big)^2\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2\\
&+2\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2.
\end{aligned}\tag{77}$$

In order to properly upper-bound the term $\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2$ in (77), we focus on the smoothness of $\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ with respect to $\boldsymbol{\theta}$ at the fixed point $\boldsymbol{\mu}_{m+1}$ and derive that

$$\begin{aligned}
&\mathcal{L}(\boldsymbol{\theta}_{m+1},\boldsymbol{\mu}_{m+1})-\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})\\
\ge&\langle\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1}),\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\rangle-\frac{L_{\theta\theta}}{2}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2,
\end{aligned}\tag{78}$$

where (78) uses the $L_{\theta\theta}$-smoothness of $\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ in Corollary 3. For $\langle\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1}),\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\rangle$ on the right hand of (78), we have that

$$\begin{aligned}
&\langle\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1}),\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\rangle\\
=&\langle\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\rangle\\
&+\langle\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})-\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}},\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\rangle\\
\ge&\frac{1}{\eta_{\theta,m}}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2-\frac{\eta_{\theta,m}}{2}\big\|\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})\\
&-\bar{\boldsymbol{g}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\big\|^2-\frac{1}{2\eta_{\theta,m}}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2
\end{aligned}\tag{79}$$

$$\begin{aligned}
\ge&\frac{1}{2\eta_{\theta,m}}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2-\frac{\eta_{\theta,m}}{2}\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2,
\end{aligned}\tag{80}$$

where (79) uses (73) and the Cauchy's inequality that $\langle a,b\rangle\ge-\frac{k}{2}\|a\|^2-\frac{1}{2k}\|y\|^2$ for all $k>0$, and (80) can be achieved by Case (ii) in Corollary 2. Substituting (80) into (78) and using $\eta_{\theta,m}=1/(2\sqrt{m}+L_{\theta\theta})$, i.e., $\frac{1}{2\eta_{\theta,m}}-\frac{L_{\theta\theta}}{2}=\sqrt{m}$, we have

$$\begin{aligned}
&\mathcal{L}(\boldsymbol{\theta}_{m+1},\boldsymbol{\mu}_{m+1})-\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})\\
\ge&\sqrt{m}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2-\frac{\eta_{\theta,m}}{2}\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2.
\end{aligned}\tag{81}$$

We introduce a auxiliary term

$$\mathcal{C}_m:=\mathcal{L}(\boldsymbol{\theta}_{m+1},\boldsymbol{\mu}_{m+2})-\mathcal{L}(\boldsymbol{\theta}_{m+1},\boldsymbol{\mu}_{m+1}).\tag{82}$$

By the definition of $\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ in (5) and its affine dependence on $\boldsymbol{\mu}$, $\mathcal{C}_m$ admits the exact expression

$$\mathcal{C}_m=\frac{1}{N}\sum_{i=1}^N(\mu_{i,m+2}-\mu_{i,m+1})\big(G_i(\boldsymbol{\theta}_{m+1})-c_i\big).\tag{83}$$

By the non-expansiveness of the projection operator and (45), we have

$$\begin{aligned}
&\big\|\boldsymbol{\mu}_{m+2}-\boldsymbol{\mu}_{m+1}\big\|\\
=&\big\|P_{\boldsymbol{\mathcal{U}}}\big(\boldsymbol{\mu}_{m+1}-\eta_{\mu,m+1}\bar{\boldsymbol{h}}_{app}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_{m+1}},\hat{\boldsymbol{\mu}}_{m+1}}\big)-P_{\boldsymbol{\mathcal{U}}}(\boldsymbol{\mu}_{m+1})\big\|\\
\le&\eta_{\mu,m+1}\sqrt{N}L_1^\mu.
\end{aligned}\tag{84}$$

Combining (83)-(84), we use Assumption 3 and further have

$$|\mathcal{C}_m|\le(R_g+c_{max})L_1^\mu\eta_{\mu,m+1}=\mathcal{O}\Big(\frac{1}{m}\Big).\tag{85}$$

By the definition of $\mathcal{C}_m$ in (82), (81) can be rewritten as

$$\begin{aligned}
&\sqrt{m}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2\\
\le&\mathcal{L}(\boldsymbol{\theta}_{m+1},\boldsymbol{\mu}_{m+2})-\mathcal{C}_m-\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})\\
&+\frac{\eta_{\theta,m}}{2}\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2.
\end{aligned}\tag{86}$$

Summing (86) over $m=1,\cdots,M-1$, we have

$$\begin{aligned}&\sum_{m=1}^{M-1}\sqrt{m}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2\\ \leq&\mathcal{L}(\boldsymbol{\theta}_M,\boldsymbol{\mu}_{M+1})-\mathcal{L}(\boldsymbol{\theta}_1,\boldsymbol{\mu}_2)-\sum_{m=1}^{M-1}\mathcal{C}_m\\ &+\sum_{m=1}^{M-1}\frac{\eta_{\theta,m}}{2}\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2\\ \leq&\frac{2(R_f+\mu_{max}R_g)}{1-\gamma}+\sum_{m=1}^{M-1}|\mathcal{C}_m|\\ &+\sum_{m=1}^{M-1}\frac{\eta_{\theta,m}}{2}\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2.\end{aligned}\tag{87}$$

Recall from Theorem 2, using $\|\hat{\theta}^i_{j,m}-\theta_{j,m}\|\sim\mathcal{O}(1/\sqrt{m})$ and $|\hat{\mu}^i_{j,m}-\mu_{j,m}|\sim\mathcal{O}(1/m)$, we have,

$$\begin{aligned}&\sum_{m=1}^{M-1}\frac{\eta_{\theta,m}}{2}\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2\\ \leq&\mathcal{O}(1)+\mathcal{O}\left(\sqrt{M}\frac{\log(2/\delta)}{2K_\theta}\right)+\mathcal{O}\Big(\sqrt{M}\epsilon_0(\kappa,\kappa_p)^2\Big).\end{aligned}\tag{88}$$

Substituting (85), (88) into (87), we obtain

$$\begin{aligned}&\sum_{m=1}^{M-1}\sqrt{m}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2\\ \leq&\mathcal{O}\big(\log M\big)+\mathcal{O}\left(\sqrt{M}\frac{\log(2/\delta)}{2K_\theta}\right)\\ &+\mathcal{O}\Big(\sqrt{M}\epsilon_0(\kappa,\kappa_p)^2\Big).\end{aligned}\tag{89}$$

Define χ_m as a weight and described as

$$\chi_m=\frac{\sqrt{m}}{2\Big(2\sqrt{m}+L_{\theta\theta}+\sqrt{N}L_1^\theta\Big)^2}\sim\mathcal{O}\Big(\frac{1}{\sqrt{m}}\Big),\tag{90}$$

then multiplying both sides of (77) by χ_m and summing over $m=1,\cdots,M-1$, we have, by (89),

$$\begin{aligned}&\sum_{m=1}^{M-1}\chi_m\big[\mathcal{Y}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})\big]^2\\ \leq&\sum_{m=1}^{M-1}\sqrt{m}\big\|\boldsymbol{\theta}_{m+1}-\boldsymbol{\theta}_m\big\|^2\\ &+2\sum_{m=1}^{M-1}\chi_m\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)^2\\ =&\mathcal{O}\big(\log M\big)+\mathcal{O}\left(\sqrt{M}\frac{\log(2/\delta)}{2K_\theta}\right)+\mathcal{O}\Big(\sqrt{M}\epsilon_0(\kappa,\kappa_p)^2\Big).\end{aligned}\tag{91}$$

Dividing both sides by $\sum_{m=1}^{M-1}\chi_m=\mathcal{O}(\sqrt{M})$, we obtain

$$\begin{aligned}&\frac{1}{\sum_{m=1}^{M-1}\chi_m}\sum_{m=1}^{M-1}\chi_m\big[\mathcal{Y}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})\big]^2\\ =&\mathcal{O}\Big(\frac{\log M}{\sqrt{M}}\Big)+\mathcal{O}\left(\frac{\log(2/\delta)}{2K_\theta}\right)+\mathcal{O}\Big(\epsilon_0(\kappa,\kappa_p)^2\Big).\end{aligned}\tag{92}$$

Since $\frac{1}{\sum_{m=1}^{M-1}\chi_m}\sum_{m=1}^{M-1}\chi_m\big[\mathcal{Y}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})\big]^2$ is a weighted average of $M-1$ non-negative terms, there must exist $m^*\in\{1,\cdots,M-1\}$ such that

$$\begin{aligned}\big[\mathcal{Y}(\boldsymbol{\theta}_{m^*},\boldsymbol{\mu}_{m^*+1})\big]^2=&\mathcal{O}\Big(\frac{\log M}{\sqrt{M}}\Big)+\mathcal{O}\left(\frac{\log(2/\delta)}{2K_\theta}\right)\\ &+\mathcal{O}\Big(\epsilon_0(\kappa,\kappa_p)^2\Big).\end{aligned}\tag{93}$$

By substituting the settings $M=\tilde{\mathcal{O}}(1/\epsilon^2)$ and $K_\theta=\log(2/\delta)/2\epsilon$ into (93), we can rigorously complete the proof. □

IX. Appendix

In this section, we provide detailed proofs of the lemmas presented in the main body and the convergence analysis.

A. Proof of Lemma 1

Proof. Recalling the definition of $\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ provided in (5), the expression of gradient $\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$ in (15) can be directly derived.

For $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})$, we combine the global Lagrangian Q-function in (10) with the policy gradient theorem in [29] and have

$$\begin{aligned}&\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\mu})\\ =&\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_\rho^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[Q^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})\nabla_{\theta_i}\log\boldsymbol{\pi_\theta}(\boldsymbol{a}|\boldsymbol{s})\Big]\\ =&\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_\rho^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[Q^{\boldsymbol{\pi_\theta}}(\boldsymbol{s},\boldsymbol{a};\boldsymbol{\mu})\\ &\times\nabla_{\theta_i}\Big(\sum_{j=1}^{N}\log\pi_j(a_j|s_{\mathcal{N}^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}^{E,\kappa_p}_{j,-j}})\Big)\Big].\end{aligned}\tag{94}$$

Since only the policies $\{\pi_j(\cdot|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\}_{j\in\mathcal{N}_i^{E,\kappa_p}}$ are related to θ_i, (94) can consequently derive (16). □

B. Proof of Lemma 2

Proof. In the CMARL problem, for any joint policy $\boldsymbol{\pi_\theta}$, let $\xi_{1,t}=\Pr^{\boldsymbol{\pi_\theta}}\big(s_{i,t},a_{i,t}|\boldsymbol{s}_0=(s_{\mathcal{N}_i^{E,\kappa}},s_{\mathcal{N}_{-i}^{E,\kappa}}),\boldsymbol{a}_0=(a_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}})\big)$ represent the probability of occurrence of $(s_{i,t},a_{i,t})$ at time t under joint policy $\boldsymbol{\pi_\theta}$, conditioned on the initial state $(s_{\mathcal{N}_i^{E,\kappa}},s_{\mathcal{N}_{-i}^{E,\kappa}})$ and initial action $(a_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}})$. Similarly, define $\xi_{2,t}=\Pr^{\boldsymbol{\pi_\theta}}\big(s_{i,t},a_{i,t}|\boldsymbol{s}_0=(s_{\mathcal{N}_i^{E,\kappa}},s'_{\mathcal{N}_{-i}^{E,\kappa}}),\boldsymbol{a}_0=(a_{\mathcal{N}_i^{E,\kappa}},a'_{\mathcal{N}_{-i}^{E,\kappa}})\big)$. Let $\mathrm{TV}(\xi_{1,t},\xi_{2,t})$ represent the total variation distance between $\xi_{1,t}$ and $\xi_{2,t}$. From the definitions of the state transition probability functions $\{\mathcal{P}_i(s'_i|s_{\mathcal{N}_i^E},a_i)\}_{i\in\mathcal{N}}$ and coupled policies $\{\pi_i(\cdot|s_{\mathcal{N}_i^{E,\kappa_p}},\theta_i,\theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})\}_{i\in\mathcal{N}}$, we can directly have that $\xi_{1,t}=\xi_{2,t}$ when $t\leq h(\kappa,\kappa_p)$. Given this fact, we obtain

$$\begin{aligned}&\Big|Q_i^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},s_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}},a_{\mathcal{N}_{-i}^{E,\kappa}};\boldsymbol{\mu})\\ &-Q_i^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_i^{E,\kappa}},s'_{\mathcal{N}_{-i}^{E,\kappa}},a_{\mathcal{N}_i^{E,\kappa}},a'_{\mathcal{N}_{-i}^{E,\kappa}};\boldsymbol{\mu})\Big|\end{aligned}$$

$$=\Big|\frac{1}{N}\sum_{t=0}^{\infty}\gamma^t\mathbb{E}_{\xi_{1,t}}[f_i(s_{i,t},a_{i,t})+\mu_i g_i(s_{i,t},a_{i,t})] - \frac{1}{N}\sum_{t=0}^{\infty}\gamma^t\mathbb{E}_{\xi_{2,t}}[f_i(s_{i,t},a_{i,t})+\mu_i g_i(s_{i,t},a_{i,t})]\Big| \quad (95)$$

$$\leq\frac{1}{N}\sum_{t=0}^{\infty}\Big|\gamma^t\mathbb{E}_{\xi_{1,t}}[r_i(s_{i,t},a_{i,t})+\mu_i g_i(s_{i,t},a_{i,t})] - \gamma^t\mathbb{E}_{\xi_{2,t}}[r_i(s_{i,t},a_{i,t})+\mu_i g_i(s_{i,t},a_{i,t})]\Big| \quad (96)$$

$$=\frac{1}{N}\sum_{t=h(\kappa,\kappa_p)+1}^{\infty}\Big|\gamma^t\mathbb{E}_{\xi_{1,t}}[r_i(s_{i,t},a_{i,t})+\mu_i g_i(s_{i,t},a_{i,t})] - \gamma^t\mathbb{E}_{\xi_{2,t}}[r_i(s_{i,t},a_{i,t})+\mu_i g_i(s_{i,t},a_{i,t})]\Big| \quad (97)$$

$$\leq\sum_{t=h(\kappa,\kappa_p)+1}^{\infty}2\gamma^t(R_f+\mu_{max}R_g)\mathrm{TV}(\xi_{1,t},\xi_{2,t}) \quad (98)$$

$$\leq\frac{2(R_f+\mu_{max}R_g)}{(1-\gamma)\gamma^{\kappa-h(\kappa,\kappa_p)}}\gamma^{\kappa+1}, \quad (99)$$

where (95) comes from the definition of local Lagrangian Q-function in (11), (96) can be obtained by the fact that $|\sum_{t=0}^{\infty}(x_t-y_t)|\leq\sum_{t=0}^{\infty}|x_t-y_t|,\forall x_t,y_t\in\mathbb{R}$, (97) is achieved by $\xi_{1,t}=\xi_{2,t}$ for all $t\leq h(\kappa,\kappa_p)$, (98) follows from Assumptions 3 and 1, and (99) comes from the fact that $\mathrm{TV}(\xi_{1,t},\xi_{2,t})$ is upper bounded by 1. □

C. Proof of Lemma 3

Proof. For any $l\in\mathcal{N}_{-i}^{E,\kappa+2\kappa_p}$, we have

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[Q_{tru,l}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_l^{E,\kappa}},a_{\mathcal{N}_l^{E,\kappa}};\boldsymbol{\mu})\times\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big]\\
=&\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}},\boldsymbol{a}\sim\boldsymbol{\pi_\theta}}\Big[Q_{tru,l}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_l^{E,\kappa}},a_{\mathcal{N}_l^{E,\kappa}};\boldsymbol{\mu})\times\nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big]\\
=&\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\sum_{\boldsymbol{s},\boldsymbol{a}}d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s})\prod_{i=1}^{N}\pi_i(a_i|s_{\mathcal{N}_i^{E,\kappa_p}},\theta_i,\theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})\times\Big[Q_{tru,l}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_l^{E,\kappa}},a_{\mathcal{N}_l^{E,\kappa}};\boldsymbol{\mu})\frac{\nabla_{\theta_i}\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})}{\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})}\Big]\\
=&\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\sum_{\boldsymbol{s},a_{-j}}d_{\boldsymbol{\rho}}^{\boldsymbol{\pi_\theta}}(\boldsymbol{s})\prod_{i\neq j}\pi_i(a_i|s_{\mathcal{N}_i^{E,\kappa_p}},\theta_i,\theta_{\mathcal{N}_{i,-i}^{E,\kappa_p}})\times\Big[Q_{tru,l}^{\boldsymbol{\pi_\theta}}(s_{\mathcal{N}_l^{E,\kappa}},a_{\mathcal{N}_l^{E,\kappa}};\boldsymbol{\mu})\sum_{a_j}\nabla_{\theta_i}\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\\
&\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\Big]\\
=&0,
\end{aligned} \quad (100)$$

where the last inequality can be obtained by the fact that

$$\begin{aligned}
&\sum_{a_j}\nabla_{\theta_i}\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})\\
=&\nabla_{\theta_i}\sum_{a_j}\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}},\theta_j,\theta_{\mathcal{N}_{j,-j}^{E,\kappa_p}})=\nabla_{\theta_i}1=0.
\end{aligned} \quad (101)$$

□

D. Proof of Lemma 5

Proof. For Case (i), by the definition of $\hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k)$ in (20), we have

$$\begin{aligned}
&\mathbb{E}_{T_1(k)}\Big[\hat{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_m}(k)\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\
=&\frac{1}{N}\Big(\mathbb{E}_{T_1(k)}\Big[\sum_{t=0}^{T_1(k)}\gamma^{t/2}g_{i,t}\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]-c_i\Big)\\
=&\frac{1}{N}\Big(\mathbb{E}_{T_1(k)}\Big[\sum_{t=0}^{\infty}\mathbb{1}_{\{0\leq t\leq T_1(k)\}}\gamma^{t/2}g_{i,t}\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]-c_i\Big).
\end{aligned} \quad (102)$$

By the boundedness of rewards in Assumption 3, for any $T'>0$, we have that

$$\begin{aligned}
&\mathbb{E}_{T_1(k)}\Big[\sum_{t=0}^{T'}\mathbb{1}_{\{0\leq t\leq T_1(k)\}}\gamma^{t/2}g_{i,t}\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\
\leq&R_g\mathbb{E}_{T_1(k)}\Big[\sum_{t=0}^{T'}\mathbb{1}_{\{0\leq t\leq T_1(k)\}}\gamma^{t/2}\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big].
\end{aligned} \quad (103)$$

Note that on the right-hand side of (103), the random variable in the expectation is monotonically increasing and the limit as $T'\to\infty$ exists. By using the monotone convergence theorem in [31], we can have that

$$\begin{aligned}
&\frac{1}{N}\mathbb{E}_{T_1(k)}\Big[\sum_{t=0}^{\infty}\mathbb{1}_{\{0\leq t\leq T_1(k)\}}\gamma^{t/2}g_{i,t}\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\
=&\frac{1}{N}\sum_{t=0}^{\infty}\mathbb{E}\Big[\mathbb{E}_{T_1(k)}[\mathbb{1}_{\{0\leq t\leq T_1(k)\}}]\gamma^{t/2}g_{i,t}\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\
=&\frac{1}{N}\sum_{t=0}^{\infty}\mathbb{E}\Big[\gamma^t g_{i,t}\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\
=&\frac{1}{N}\mathbb{E}\Big[\sum_{t=0}^{\infty}\gamma^t g_{i,t}\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big],
\end{aligned} \quad (104)$$

where the second equality can be achieved by the fact that $T_1(k)\sim\mathrm{Geom}(1-\gamma^{1/2})$ and $\mathbb{E}_{T_1(k)}[\mathbb{1}_{\{0\leq t\leq T_1(k)\}}]=\gamma^{t/2}$, and the last equality uses the dominated convergence theorem. By substituting (104) into (102) and referring to (68), Case (i) can be rigorously proved.

For Case (ii), by the definition of $\hat{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}(k)$ in (24) and the independence of $T_2(k)$ and $T_3(k)$, we have

$$\begin{aligned}
&\mathbb{E}_{T_2(k),T_3(k)}\Big[\hat{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}(k)\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\
=&\frac{1}{1-\gamma}\mathbb{E}_{T_2(k),T_3(k)}\Big[\hat{Q}_{i,T_2(k)}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\Big(\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_{j,T_2(k)}|\\
&s_{\mathcal{N}_j^{E,\kappa_p},T_2(k)},\hat{\theta}_{j,m}^i,\hat{\theta}_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}^i)\Big)\Big|\boldsymbol{s}_0\sim\boldsymbol{\rho},\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\
=&\frac{1}{1-\gamma}\mathbb{E}_{T_2(k)}\Big[\mathbb{E}_{T_3(k)}\Big[\hat{Q}_{i,T_2(k)}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\\
&\times\Big(\sum_{j\in\mathcal{N}_i^{E,\kappa_p}}\nabla_{\theta_i}\log\pi_j(a_{j,T_2(k)}|s_{\mathcal{N}_j^{E,\kappa},T_2(k)},\hat{\theta}_{j,m}^i,\hat{\theta}_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}^i)\Big)\Big|
\end{aligned}$$

□

$$\boldsymbol{s}_{T_2(k)}, \boldsymbol{a}_{T_2(k)}, \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\Big|\boldsymbol{s}_0 \sim \boldsymbol{\rho}, \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]. \tag{105}$$

According to Theorem 3.4 in [32], we have that

$$\begin{aligned}&\widehat{Q_i^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}}(\boldsymbol{s}_{T_2(k)}, \boldsymbol{a}_{T_2(k)}; \hat{\boldsymbol{\mu}}_{m+1}^i)\\ =&\mathbb{E}_{T_3(k)}\Big[\hat{Q}_{i,T_2(k)}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_{m+1}}\Big|\boldsymbol{s}_{T_2(k)}, \boldsymbol{a}_{T_2(k)}, \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big].\end{aligned} \tag{106}$$

Substituting (106) into (105), we further have

$$\begin{aligned}&\mathbb{E}_{T_2(k),T_3(k)}\Big[\hat{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}(k)\Big|\boldsymbol{s}_0 \sim \boldsymbol{\rho}, \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\ =&\frac{1}{1-\gamma}\mathbb{E}_{T_2(k)}\Big[\widehat{Q_i^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}}(\boldsymbol{s}_{T_2(k)}, \boldsymbol{a}_{T_2(k)}; \hat{\boldsymbol{\mu}}_{m+1}^i)\\ &\times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i}\log\pi_j(a_{j,T_2(k)}|s_{\mathcal{N}_j^{E,\kappa_p},T_2(k)},\\ &\hat{\theta}_{j,m}^i, \hat{\theta}_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}^i)\Big|\boldsymbol{s}_0 \sim \boldsymbol{\rho}, \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\ =&\frac{1}{1-\gamma}\mathbb{E}_{T_2(k)}\Big[\sum_{t'=0}^{\infty}\mathbb{1}_{\{t'=T_2(k)\}}\widehat{Q_i^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}}(\boldsymbol{s}_{T_2(k)}, \boldsymbol{a}_{T_2(k)}; \hat{\boldsymbol{\mu}}_{m+1}^i)\\ &\times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i}\log\pi_j(a_{j,T_2(k)}|s_{\mathcal{N}_j^{E,\kappa_p},T_2(k)},\\ &\hat{\theta}_{j,m}^i, \hat{\theta}_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}^i)\Big|\boldsymbol{s}_0 \sim \boldsymbol{\rho}, \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\ =&\frac{1}{1-\gamma}\sum_{t'=0}^{\infty}\mathbb{P}\big(t'=T_2(k)\big)\mathbb{E}\Big[\widehat{Q_i^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}}(\boldsymbol{s}_{t'}, \boldsymbol{a}_{t'}; \hat{\boldsymbol{\mu}}_{m+1}^i)\\ &\times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i}\log\pi_j(a_{j,t'}|s_{\mathcal{N}_j^{E,\kappa},t'}, \hat{\theta}_{j,m}^i, \hat{\theta}_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}^i)\Big|\\ &\boldsymbol{s}_0 \sim \boldsymbol{\rho}, \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\\ =&\sum_{t'=0}^{\infty}\gamma^{t'}\mathbb{E}\Big[\widehat{Q_i^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}}(\boldsymbol{s}_{t'}, \boldsymbol{a}_{t'}; \hat{\boldsymbol{\mu}}_{m+1}^i)\\ &\times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i}\log\pi_j(a_{j,t'}|s_{\mathcal{N}_j^{E,\kappa_p},t'}, \hat{\theta}_{j,m}^i, \hat{\theta}_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}^i)\Big|\\ &\boldsymbol{s}_0 \sim \boldsymbol{\rho}, \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}\Big]\end{aligned} \tag{107}$$

$$\begin{aligned}=&\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}, \boldsymbol{a}\sim\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}\Big[\widehat{Q_i^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}}(\boldsymbol{s}, \boldsymbol{a}; \hat{\boldsymbol{\mu}}_{m+1}^i)\\ &\times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i}\log\pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}}, \hat{\theta}_{j,m}^j, \hat{\theta}_{\mathcal{N}_{j,-j}^{E,\kappa_p},t}^j)\Big]\end{aligned} \tag{108}$$

$$=g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_{m+1}},$$

where the equality (107) comes from the fact that $T_2(k) \sim \mathrm{Geom}(1-\gamma)$ and $\mathbb{P}\big(t'=T_2(k)\big) = (1-\gamma)\gamma^{t'}$, (108) comes from the definition of the discounted state visitation distribution in (14), and the last equality achieves by the definition of $g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_{m+1}}$ in (70). Hence, the proof of Case (ii) is completed. □

E. Proof of Lemma 6

Proof. For Case (i), by the definitions of $\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\mu})$ in (15) and $G_i(\boldsymbol{\theta})$ in (3), we have

$$\begin{aligned}&\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_m)\\ =&\frac{1}{N}\mathbb{E}_{\boldsymbol{s}\sim\boldsymbol{\rho}}\Big[\sum_{t=0}^{\infty}\gamma^t g_i(s_{i,t}, a_{i,t})\Big|\boldsymbol{s}_0 = \boldsymbol{s}, \boldsymbol{a}_t \sim \boldsymbol{\pi}_{\boldsymbol{\theta}_m}(\cdot|\boldsymbol{s}_t)\Big]\\ =&\frac{1}{N}\sum_{t=0}^{\infty}\gamma^t\int_{\boldsymbol{s}_0,\boldsymbol{a}_0} g_i(s_{i,t}, a_{i,t})\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})d_{\boldsymbol{s}_{0:t},\boldsymbol{a}_{0:t}},\end{aligned} \tag{109}$$

where $\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})$ is the probability of the sequence $\{\boldsymbol{s}, \boldsymbol{a}\}_{0:t}$ generated by the joint policy $\boldsymbol{\pi}_{\boldsymbol{\theta}_m}$, which is described as

$$\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t}) = \prod_{h=0}^{t-1}\boldsymbol{\mathcal{P}}(\boldsymbol{s}_{h+1}|\boldsymbol{s}_h, \boldsymbol{a}_h)\prod_{h=0}^{t}\boldsymbol{\pi}_{\boldsymbol{\theta}}(\boldsymbol{a}_h|\boldsymbol{s}_h)\boldsymbol{\rho}(\boldsymbol{s}_0). \tag{110}$$

Similar to (110), define $\rho_t^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})$ as the probability of the sequence $\{\boldsymbol{s}, \boldsymbol{a}\}_{0:t}$ generated by the joint policy $\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$. By using (109), we have that

$$\begin{aligned}&\Big\|\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_m) - \nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\mu}}_m)\Big\|\\ =&\frac{1}{N}\Big|\sum_{t=0}^{\infty}\gamma^t\int_{\boldsymbol{s}_0,\boldsymbol{a}_0} g_i(s_{i,t}, a_{i,t})\\ &\times\big(\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t}) - \rho_t^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})\big)d_{\boldsymbol{s}_{0:t},\boldsymbol{a}_{0:t}}\Big|\\ \le&\frac{1}{N}\sum_{t=0}^{\infty}\gamma^t\int_{\boldsymbol{s}_0,\boldsymbol{a}_0}\big|g_i(s_{i,t}, a_{i,t})\big|\\ &\times\big|\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t}) - \rho_t^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})\big|d_{\boldsymbol{s}_{0:t},\boldsymbol{a}_{0:t}},\end{aligned} \tag{111}$$

where the equality uses the fact that Lagrangian gradient $\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\mu})$ is real value and the last inequality can be achieved by the absolute value inequality, i.e., $|x+y| \le |x| + |y|$ for all $x, y \in \mathbb{R}$. Based on the definition of $\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})$ in (110), the term $\big|\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t}) - \rho_t^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})\big|$ in the right-hand of (111) can be further written as

$$\begin{aligned}&\big|\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t}) - \rho_t^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})\big|\\ =&\Big|\boldsymbol{\rho}(\boldsymbol{s}_0)\prod_{h=0}^{t-1}\boldsymbol{\mathcal{P}}(\boldsymbol{s}_{h+1}|\boldsymbol{s}_h, \boldsymbol{a}_h)\\ &\times\Big(\prod_{h=0}^{t}\boldsymbol{\pi}_{\boldsymbol{\theta}_m}(\boldsymbol{a}_h|\boldsymbol{s}_h) - \prod_{h=0}^{t}\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}(\boldsymbol{a}_h|\boldsymbol{s}_h)\Big)\Big|\\ =&\boldsymbol{\rho}(\boldsymbol{s}_0)\prod_{h=0}^{t-1}\boldsymbol{\mathcal{P}}(\boldsymbol{s}_{h+1}|\boldsymbol{s}_h, \boldsymbol{a}_h)\\ &\times\Big|\prod_{h=0}^{t}\hat{\boldsymbol{\pi}}_{\mathbf{1}_N\otimes\boldsymbol{\theta}_m}(\boldsymbol{a}_h|\boldsymbol{s}_h) - \prod_{h=0}^{t}\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}(\boldsymbol{a}_h|\boldsymbol{s}_h)\Big|\end{aligned} \tag{112}$$

$$\begin{aligned}=&\boldsymbol{\rho}(\boldsymbol{s}_0)\prod_{h=0}^{t-1}\boldsymbol{\mathcal{P}}(\boldsymbol{s}_{h+1}|\boldsymbol{s}_h, \boldsymbol{a}_h)\Big|(\mathbf{1}_N\otimes\boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m)^\top\\ &\Big(\sum_{h'=0}^{t}\nabla_{\hat{\boldsymbol{\theta}}}\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}}(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'})\prod_{\substack{h=0\\h\neq h'}}^{t}\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}}(\boldsymbol{a}_h|\boldsymbol{s}_h)\Big)\Big|\end{aligned} \tag{113}$$

$$\begin{aligned}=&\boldsymbol{\rho}(\boldsymbol{s}_0)\prod_{h=0}^{t-1}\boldsymbol{\mathcal{P}}(\boldsymbol{s}_{h+1}|\boldsymbol{s}_h, \boldsymbol{a}_h)\Big|(\mathbf{1}_N\otimes\boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m)^\top\\ &\Big(\sum_{h'=0}^{t}\nabla_{\hat{\boldsymbol{\theta}}}\log\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}}(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'})\prod_{h=0}^{t}\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}}(\boldsymbol{a}_h|\boldsymbol{s}_h)\Big)\Big|\end{aligned}$$

$$\leq \|\mathbf{1}_N \otimes \boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m\| \sum_{h'=0}^{t} \|\nabla_{\hat{\boldsymbol{\theta}}} \log \hat{\boldsymbol{\pi}}_{\check{\boldsymbol{\theta}}}(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'})\| \tag{114}$$

$$\leq (t+1)BN\|\mathbf{1}_N \otimes \boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m\|, \tag{115}$$

where the equality (112) uses the fact that $\hat{\boldsymbol{\pi}}_{\mathbf{1}_N \otimes \boldsymbol{\theta}_m} = \hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}$, equality (113) comes from the mean value theorem and $\check{\boldsymbol{\theta}} = \xi(\mathbf{1}_N \otimes \boldsymbol{\theta}_m) + (1-\xi)\hat{\boldsymbol{\theta}}_m$ with $\xi \in [0,1]$, the inequality (114) uses the fact that $|x^\top y| \leq \|x\|\|y\|$ for all $x, y \in \mathbb{R}^{\sum_{i=1}^N d_{\theta_i}}$, and the last inequality can be obtained by Assumption 5 and

$$\begin{aligned}
&\|\nabla_{\hat{\boldsymbol{\theta}}} \log \hat{\boldsymbol{\pi}}_{\check{\boldsymbol{\theta}}}(\boldsymbol{a}_{h'}|\boldsymbol{s}_{h'})\| \\
=&\left\| \begin{array}{c} \nabla_{\hat{\theta}_1^1} \log \pi_1(a_{1,h'}|s_{\mathcal{N}_1^{E,\kappa_p},h'}, \check{\theta}_1^1, \check{\theta}^1_{\mathcal{N}_{1,-1}^{E,\kappa_p}}) \\ \vdots \\ \nabla_{\hat{\theta}_N^1} \log \pi_1(a_{1,h'}|s_{\mathcal{N}_1^{E,\kappa_p},h'}, \check{\theta}_1^1, \check{\theta}^1_{\mathcal{N}_{1,-1}^{E,\kappa_p}}) \\ \vdots \\ \nabla_{\hat{\theta}_N^N} \log \pi_N(a_{N,h'}|s_{\mathcal{N}_N^{E,\kappa_p},h'}, \check{\theta}_N^N, \check{\theta}^N_{\mathcal{N}_{N,-N}^{E,\kappa_p}}) \end{array} \right\| \leq BN.
\end{aligned}$$

Substituting (115) into (111), we have

$$\begin{aligned}
&\left\|\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_m) - \nabla_{\mu_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\mu}}_m)\right\| \\
\leq& \frac{1}{N}\sum_{t=0}^{\infty} \gamma^t (t+1) BNR_g \|\mathbf{1}_N \otimes \boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m\| \\
\leq& \frac{BR_g}{(1-\gamma)^2}\|\mathbf{1}_N \otimes \boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m\|,
\end{aligned}$$

which completes the Case (i).

For Case (ii), by the definitions of $\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\mu})$ in (16) and $\rho_t^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t}, \boldsymbol{a}_{0:t})$ in (110), we have

$$\begin{aligned}
&\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_{m+1}) \\
=&\frac{1}{1-\gamma}\mathbb{E}_{\boldsymbol{s}\sim d_{\boldsymbol{\rho}}^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}, \boldsymbol{a}\sim\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}\Big[Q^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}, \boldsymbol{a}; \boldsymbol{\mu}_{m+1}) \\
&\times \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \nabla_{\theta_i} \log \pi_j(a_j|s_{\mathcal{N}_j^{E,\kappa_p}}, \theta_{j,m}, \theta_{\mathcal{N}_{j,-j}^{E,\kappa_p},m})\Big] \\
=& \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \sum_{t'=0}^{\infty}\sum_{t=0}^{\infty} \gamma^{t'+t} \int_{\boldsymbol{s}_{t'},\boldsymbol{a}_{t'}} \Big(\frac{1}{N} \\
&\times \sum_{k\in\mathcal{N}} \big(r_k(s_{k,t'+t}, a_{k,t'+t}) + \mu_{k,m+1} g_k(s_{k,t'+t}, a_{k,t'+t})\big)\Big) \\
&\times \nabla_{\theta_i} \log \pi_j(a_{j,t'}|s_{\mathcal{N}_j^{E,\kappa_p},t'}, \theta_{j,m}, \theta_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}) \\
&\times \rho_{t'+t}^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t'+t}, \boldsymbol{a}_{0:t'+t}) d_{\boldsymbol{s}_{0:t'+t}} d_{\boldsymbol{a}_{0:t'+t}}.
\end{aligned} \tag{116}$$

Based on the definition of $\nabla_{\theta_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\mu}}_{m+1})$ in (69), we have

$$\begin{aligned}
&\|\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_{m+1}) - \nabla_{\theta_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m, \hat{\boldsymbol{\mu}}_{m+1})\| \\
=&\Bigg\| \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \sum_{t'=0}^{\infty}\sum_{t=0}^{\infty} \gamma^{t'+t} \Bigg(\int_{\boldsymbol{s}_{t'},\boldsymbol{a}_{t'}} \Big(\frac{1}{N} \\
&\times \sum_{k\in\mathcal{N}} \big(\mu_{k,m+1} g_k(s_{k,t'+t}, a_{k,t'+t}) \\
&- \hat{\mu}_{k,m+1}^i g_k(s_{k,t'+t}, a_{k,t'+t})\big)\Big) \\
&\times \nabla_{\theta_i} \log \pi_j(a_{j,t'}|s_{\mathcal{N}_j^{E,\kappa_p},t'}, \theta_{j,m}, \theta_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}) \\
&\times \rho_{t'+t}^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t'+t}, \boldsymbol{a}_{0:t'+t}) d_{\boldsymbol{s}_{0:t'+t}} d_{\boldsymbol{a}_{0:t'+t}} \Bigg) \\
&+ \gamma^{t'+t} \Bigg(\int_{\boldsymbol{s}_{t'},\boldsymbol{a}_{t'}} \Big(\frac{1}{N} \sum_{k\in\mathcal{N}} \big(r_k(s_{k,t'+t}, a_{k,t'+t}) \\
&+ \hat{\mu}_{k,m+1}^i g_k(s_{k,t'+t}, a_{k,t'+t})\big)\Big) \\
&\times \Big(\nabla_{\theta_i} \log \pi_j(a_{j,t'}|s_{\mathcal{N}_j^{E,\kappa_p},t'}, \theta_{j,m}, \theta_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}) \\
&- \nabla_{\theta_i} \log \pi_j(a_{j,t'}|s_{\mathcal{N}_j^{E,\kappa_p},t'}, \hat{\theta}_{j,m}^i, \hat{\theta}^i_{\mathcal{N}_{j,-j}^{E,\kappa_p},m})\Big) \\
&\times \rho_{t'+t}^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t'+t}, \boldsymbol{a}_{0:t'+t}) d_{\boldsymbol{s}_{0:t'+t}} d_{\boldsymbol{a}_{0:t'+t}} \Bigg) \\
&+ \gamma^{t'+t} \Bigg(\int_{\boldsymbol{s}_{t'},\boldsymbol{a}_{t'}} \Big(\frac{1}{N} \sum_{k\in\mathcal{N}} \big(r_k(s_{k,t'+t}, a_{k,t'+t}) \\
&+ \hat{\mu}_{k,m+1}^i g_k(s_{k,t'+t}, a_{k,t'+t})\big)\Big) \\
&\times \nabla_{\theta_i} \log \pi_j(a_{j,t'}|s_{j,t'}, \hat{\theta}_{j,m}^i, \hat{\theta}^i_{\mathcal{N}_{j,-j}^{E,\kappa_p},m}) \\
&\times \big(\rho_{t'+t}^{\boldsymbol{\pi}_{\boldsymbol{\theta}_m}}(\boldsymbol{s}_{0:t'+t}, \boldsymbol{a}_{0:t'+t}) \\
&- \rho_{t'+t}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_t}}(\boldsymbol{s}_{0:t'+t}, \boldsymbol{a}_{0:t'+t})\big) d_{\boldsymbol{s}_{0:t'+t}} d_{\boldsymbol{a}_{0:t'+t}} \Bigg) \Bigg\| \\
\leq& \sum_{j\in\mathcal{N}_i^{E,\kappa_p}} \sum_{t'=0}^{\infty}\sum_{t=0}^{\infty} \gamma^{t'+t} \Big(\frac{BR_g}{N}\|\boldsymbol{\mu}_{m+1} - \hat{\boldsymbol{\mu}}_{m+1}^i\|_1 \\
&+ L(R_f + \tilde{\mu}_{max}R_g)\|\boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m^i\| \\
&+ (t'+t+1)B^2N(R_f + \tilde{\mu}_{max}R_g)\|\mathbf{1}_N \otimes \boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m\|\Big)
\end{aligned} \tag{117}$$

$$\begin{aligned}
\leq& \frac{B\sqrt{N}R_g}{(1-\gamma)^2}\|\boldsymbol{\mu}_{m+1} - \hat{\boldsymbol{\mu}}_{m+1}^i\| + \Big(\frac{LN(R_f + \tilde{\mu}_{max}R_g)}{(1-\gamma)^2} \\
&+ \frac{(1+\gamma)B^2N^2(R_f + \tilde{\mu}_{max}R_g)}{(1-\gamma)^3}\Big)\|\mathbf{1}_N \otimes \boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m\|,
\end{aligned} \tag{118}$$

where the inequality (117) can be obtained by Assumptions 3 and 5 and (118) uses the fact that $\|x\|_1 \leq \sqrt{N}\|x\|$ for any vector $x \in \mathbb{R}^N$. Hence, we complete the proof of this lemma. □

F. Proof of Corollary 2

Proof. For Case (i), we have that

$$\begin{aligned}
&\left\|\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_m) - \bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_m}\right\| \\
\leq& \left\|\nabla_{\mu_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_m) - \nabla_{\mu_i}\tilde{\mathcal{L}}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_m)\right\| \\
&+ \left\|\nabla_{\mu_i}\tilde{\mathcal{L}}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_m) - \bar{h}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_m}\right\| \\
\leq& L_2^\mu \left\|\mathbf{1}_N \otimes \boldsymbol{\theta}_m - \hat{\boldsymbol{\theta}}_m\right\| + 2L_1^\mu \sqrt{\frac{\log(2/\delta)}{2K_\mu}}
\end{aligned} \tag{119}$$

$$= \epsilon_\mu(m, K_\mu)/\sqrt{N}, \tag{120}$$

where (119) uses Case (i) in Lemma 6 and (71).

For Case (ii), we have

$$\left\|\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m, \boldsymbol{\mu}_{m+1}) - \bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m}, \hat{\boldsymbol{\mu}}_{m+1}}\right\|$$

$$
\begin{aligned}
&\leq \underbrace{\left\|\nabla_{\theta_i}\mathcal{L}(\boldsymbol{\theta}_m,\boldsymbol{\mu}_{m+1})-\nabla_{\theta_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_{m+1})\right\|}_{\text{(i)}}\\
&\quad+\underbrace{\left\|\nabla_{\theta_i}\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_m,\hat{\boldsymbol{\mu}}_{m+1})-g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\right\|}_{\text{(ii)}}\\
&\quad+\underbrace{\left\|g_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}-\bar{g}_{app,i}^{\hat{\boldsymbol{\pi}}_{\hat{\boldsymbol{\theta}}_m},\hat{\boldsymbol{\mu}}_{m+1}}\right\|}_{\text{(iii)}}\\
&\leq L_2^{\theta\mu}\|\mathbf{1}_N\otimes\boldsymbol{\mu}_{m+1}-\hat{\boldsymbol{\mu}}_{m+1}\|+L_2^{\theta\theta}\|\mathbf{1}_N\otimes\boldsymbol{\theta}_m-\hat{\boldsymbol{\theta}}_m\|\\
&\quad+\frac{2(R_f+\tilde{\mu}_{max}R_g)BN}{(1-\gamma)^2}\gamma^{h(\kappa,\kappa_p)+1}\\
&\quad+2L_1^{\theta}\sqrt{\frac{\log(2/\delta)}{2K_\theta}} \qquad (121)\\
&=\big(\epsilon_\theta(m,K_\theta)+\epsilon_0(\kappa,\kappa_p)\big)/\sqrt{N}, \qquad (122)
\end{aligned}
$$

where (121) can be obtained the Case (ii) in Lemma 6, (29), (72), and the fact that $\left\|\boldsymbol{\mu}_m-\hat{\boldsymbol{\mu}}_m^i\right\|\leq\left\|\mathbf{1}_N\otimes\boldsymbol{\mu}_m-\hat{\boldsymbol{\mu}}_m\right\|$. □